

Supplement to Six Birds Foundations V: Endogenous Closure

A Catalog of Structural Laws for Living, Cognitive, and Social Systems

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S1 What this supplement holds

This supplement accompanies *Six Birds Foundations V: Endogenous Closure — A Catalog of Structural Laws for Living, Cognitive, and Social Systems*. The main paper states and proves every result; its numbered items are cited here by their numbers there. The supplement gives the Lean counterpart of each numbered item and the status of each entry in the theorem catalog of the development (Section S2); the fields of every certificate (Section S5); the register of non-implications with their separating examples (Section S6); the finite computations (Section S7); the life-facing synthetic data and a proposed biological protocol (Section S8); the axioms used by the Lean development (Section S9); and the commands that rebuild everything (Section S10).

Every declaration named in Sections S2 and S9 is present at the file and line given; the tables of those two sections are generated from the Lean sources.

S2 Source concordance

Table S1 lists every numbered item of the main paper with its entry, its role, the line of the theorem catalog of the development at which that entry begins, and its Lean declarations with their kind, file and line. Declaration names omit the common prefix of the library; names beginning with **Instances.** belong to the instance modules. Items argued on paper only are marked as such.

Table S1. Source concordance.

Item	Entry and role	Lean declarations
Definition 3.1	setting, definition	<code>TheoryPackage</code> (structure) <code>lean/SixBirdsFoundationsV/Definitional/ESystem.lean:16</code>
Definition 3.2	D1, definition catalog line 54	<code>Refines</code> (def) <code>lean/SixBirdsFoundationsV/Definitional/RepairJoin.lean:13</code> <code>FiberEquiv</code> (def) <code>lean/SixBirdsFoundationsV/Definitional/RepairJoin.lean:16</code> <code>repairJoin</code> (def) <code>lean/SixBirdsFoundationsV/Definitional/RepairJoin.lean:19</code>

Item	Entry and role	Lean declarations
Proposition 3.3	D1, general theorem catalog line 54	join_refines_left (theorem) lean/SixBirdsFoundationsV/Definitional/RepairJoin.lean:23 join_refines_right (theorem) lean/SixBirdsFoundationsV/Definitional/RepairJoin.lean:29 vee_greatest_lower_bound (theorem) lean/SixBirdsFoundationsV/Definitional/RepairJoin.lean:35 join_idem (theorem) lean/SixBirdsFoundationsV/Definitional/RepairJoin.lean:57 join_comm (theorem) lean/SixBirdsFoundationsV/Definitional/RepairJoin.lean:66 join_assoc (theorem) lean/SixBirdsFoundationsV/Definitional/RepairJoin.lean:75 join_well_defined (theorem) lean/SixBirdsFoundationsV/Definitional/RepairJoin.lean:42
Definition 3.4	D2, definition catalog line 56	predictiveSurplusValue (def) lean/SixBirdsFoundationsV/Definitional/PredictiveSurplus.lean:55 predictiveSurplusXi (def) lean/SixBirdsFoundationsV/Definitional/PredictiveSurplus.lean:91
Proposition 3.5	D2, conditional general theorem catalog line 56	predictiveSurplusValue_nonnegative_iff (theorem) lean/SixBirdsFoundationsV/Definitional/PredictiveSurplus.lean:58 predictiveSurplusValue_identical (theorem) lean/SixBirdsFoundationsV/Definitional/PredictiveSurplus.lean:72 predictiveSurplusValue_exact_baseline (theorem) lean/SixBirdsFoundationsV/Definitional/PredictiveSurplus.lean:78 predictiveSurplusXi_nonnegative_of_psd (theorem) lean/SixBirdsFoundationsV/Definitional/PredictiveSurplus.lean:96 predictiveSurplusXi_chain_rule_monotone (theorem) lean/SixBirdsFoundationsV/Definitional/PredictiveSurplus.lean:103 predictiveSurplusXi_same_family_saturation (theorem) lean/SixBirdsFoundationsV/Definitional/PredictiveSurplus.lean:136
Definition 3.6	D3, definition catalog line 60	FineSourceTag (inductive) lean/SixBirdsFoundationsV/Definitional/CarriedRecord.lean:15 CarriedSource (def) lean/SixBirdsFoundationsV/Definitional/CarriedRecord.lean:26 CarriedRecord (def) lean/SixBirdsFoundationsV/Definitional/CarriedRecord.lean:54 CarriedInstrument (def) lean/SixBirdsFoundationsV/Definitional/CarriedRecord.lean:89
Lemma S3.1	D3, consistency lemma catalog line 60	not_carriedSource_fallback (theorem) lean/SixBirdsFoundationsV/Definitional/CarriedRecord.lean:115 not_carriedSource_independent_pair_witness (theorem) lean/SixBirdsFoundationsV/Definitional/CarriedRecord.lean:122 not_carriedSource_simulation_trace (theorem) lean/SixBirdsFoundationsV/Definitional/CarriedRecord.lean:131 not_carriedSource_ablation_record (theorem) lean/SixBirdsFoundationsV/Definitional/CarriedRecord.lean:139 not_carriedSource_unknown (theorem) lean/SixBirdsFoundationsV/Definitional/CarriedRecord.lean:147 not_carriedSource_contradictory (theorem) lean/SixBirdsFoundationsV/Definitional/CarriedRecord.lean:154 not_carriedRecord_of_step_not_in_scope (theorem) lean/SixBirdsFoundationsV/Definitional/CarriedRecord.lean:198 not_carriedInstrument_of_incomplete_inventory (theorem) lean/SixBirdsFoundationsV/Definitional/CarriedRecord.lean:265
Definition 3.7	D4, definition catalog line 62	RepairSort (def) lean/SixBirdsFoundationsV/Definitional/ESystem.lean:13 CarriedLedger (structure) lean/SixBirdsFoundationsV/Definitional/ESystem.lean:72 RepairMove (structure) lean/SixBirdsFoundationsV/Definitional/ESystem.lean:85 ActiveCarriedInstrument (structure) lean/SixBirdsFoundationsV/Definitional/ESystem.lean:97 ESystem (structure) lean/SixBirdsFoundationsV/Definitional/ESystem.lean:177

Item	Entry and role	Lean declarations
Definition 3.8	D4, definition catalog line 62	<code>LawfulRepairStep</code> (structure) <code>lean/SixBirdsFoundationsV/Definitional/ESystem.lean:123</code> <code>ESystem.RepairStep</code> (def) <code>lean/SixBirdsFoundationsV/Definitional/ESystem.lean:198</code>
Lemma S3.2	D4, consistency lemma catalog line 62	<code>lawfulRepairStep_requires_detection</code> (theorem) <code>lean/SixBirdsFoundationsV/Definitional/ESystem.lean:319</code> <code>lawfulRepairStep_requires_gate</code> (theorem) <code>lean/SixBirdsFoundationsV/Definitional/ESystem.lean:346</code> <code>lawfulRepairStep_requires_kernel_transition</code> (theorem) <code>lean/SixBirdsFoundationsV/Definitional/ESystem.lean:373</code> <code>lawfulRepairStep_requires_reaudit</code> (theorem) <code>lean/SixBirdsFoundationsV/Definitional/ESystem.lean:400</code> <code>not_lawfulRepairStep_without_detection</code> (theorem) <code>lean/SixBirdsFoundationsV/Definitional/ESystem.lean:427</code> <code>not_lawfulRepairStep_without_gate</code> (theorem) <code>lean/SixBirdsFoundationsV/Definitional/ESystem.lean:453</code> <code>not_lawfulRepairStep_without_kernel_transition</code> (theorem) <code>lean/SixBirdsFoundationsV/Definitional/ESystem.lean:479</code> <code>not_lawfulRepairStep_without_reaudit</code> (theorem) <code>lean/SixBirdsFoundationsV/Definitional/ESystem.lean:505</code> <code>no_repairSort_p6</code> (theorem) <code>lean/SixBirdsFoundationsV/Definitional/ESystem.lean:215</code>
Definition 3.9	D5, definition catalog line 66	<code>ChallengeEpisode</code> (structure) <code>lean/SixBirdsFoundationsV/Definitional/ClosedLoopScope.lean:77</code> <code>ChallengeHistory</code> (structure) <code>lean/SixBirdsFoundationsV/Definitional/ClosedLoopScope.lean:88</code> <code>ClosedLoopScope</code> (def) <code>lean/SixBirdsFoundationsV/Definitional/ClosedLoopScope.lean:109</code>
Lemma S3.3	D5, consistency lemma catalog line 66	<code>closedLoopScope_complete</code> (theorem) <code>lean/SixBirdsFoundationsV/Definitional/ClosedLoopScope.lean:128</code> <code>closedLoopScope_entry_carried</code> (theorem) <code>lean/SixBirdsFoundationsV/Definitional/ClosedLoopScope.lean:146</code> <code>not_closedLoopScope_of_listed_source_fallback</code> (theorem) <code>lean/SixBirdsFoundationsV/Definitional/ClosedLoopScope.lean:474</code> <code>not_closedLoopScope_of_listed_source_unknown</code> (theorem) <code>lean/SixBirdsFoundationsV/Definitional/ClosedLoopScope.lean:496</code> <code>not_closedLoopScope_of_listed_source_contradictory</code> (theorem) <code>lean/SixBirdsFoundationsV/Definitional/ClosedLoopScope.lean:518</code> <code>not_closedLoopScope_of_listed_generatedByS_false</code> (theorem) <code>lean/SixBirdsFoundationsV/Definitional/ClosedLoopScope.lean:540</code> <code>not_closedLoopScope_of_listed_inScope_false</code> (theorem) <code>lean/SixBirdsFoundationsV/Definitional/ClosedLoopScope.lean:562</code> <code>fallback_entry_cannot_borrow_carriedness</code> (theorem) <code>lean/SixBirdsFoundationsV/Definitional/ClosedLoopScope.lean:584</code>
Definition 3.10	D6, definition catalog line 68	<code>ProbeEconomy</code> (structure) <code>lean/SixBirdsFoundationsV/Definitional/ProbeEconomy.lean:45</code> <code>SameFamilySaturated</code> (def) <code>lean/SixBirdsFoundationsV/Definitional/ProbeEconomy.lean:30</code> <code>AcquisitionStrict</code> (def) <code>lean/SixBirdsFoundationsV/Definitional/ProbeEconomy.lean:34</code> <code>LawfulAllocation</code> (structure) <code>lean/SixBirdsFoundationsV/Definitional/ProbeEconomy.lean:117</code> <code>LawfulAcquisition</code> (structure) <code>lean/SixBirdsFoundationsV/Definitional/ProbeEconomy.lean:157</code> <code>LawfulRetirement</code> (structure) <code>lean/SixBirdsFoundationsV/Definitional/ProbeEconomy.lean:198</code>

Item	Entry and role	Lean declarations
Lemma S3.4	D6, consistency lemma catalog line 68	<code>lawfulAllocation_no_new_active_probe</code> (theorem) <code>lean/SixBirdsFoundationsV/Definitional/ProbeEconomy.lean:290</code> <code>lawfulAcquisition_requires_acquisitionStrict</code> (theorem) <code>lean/SixBirdsFoundationsV/Definitional/ProbeEconomy.lean:228</code> <code>lawfulRetirement_requires_active</code> (theorem) <code>lean/SixBirdsFoundationsV/Definitional/ProbeEconomy.lean:313</code> <code>lawfulRetirement_probe_absent_after</code> (theorem) <code>lean/SixBirdsFoundationsV/Definitional/ProbeEconomy.lean:333</code> <code>not_lawfulAcquisition_of_sameFamilySaturated</code> (theorem) <code>lean/SixBirdsFoundationsV/Definitional/ProbeEconomy.lean:249</code> <code>xi_sameFamilySaturated_not_acquisitionStrict</code> (theorem) <code>lean/SixBirdsFoundationsV/Definitional/ProbeEconomy.lean:395</code>
Definition 4.1	E3, E2, instance data catalog line 133	<code>Instances.eventTheory</code> (def) <code>lean/SixBirdsFoundationsV/Instances/E3Executed.lean:6</code> <code>Instances.eventPolicy</code> (def) <code>lean/SixBirdsFoundationsV/Instances/E3Executed.lean:34</code> <code>Instances.eventInstrument</code> (def) <code>lean/SixBirdsFoundationsV/Instances/E3Executed.lean:48</code> <code>Instances.eventSystem</code> (def) <code>lean/SixBirdsFoundationsV/Instances/E3Executed.lean:61</code>
Proposition 4.2	D4, instance catalog line 62	<code>Instances.eventAt</code> (theorem) <code>lean/SixBirdsFoundationsV/Instances/E3Executed.lean:38</code> <code>Instances.eventCarried</code> (theorem) <code>lean/SixBirdsFoundationsV/Instances/E3Executed.lean:44</code> <code>Instances.eventRepair</code> (theorem) <code>lean/SixBirdsFoundationsV/Instances/E3Executed.lean:81</code>
Definition 4.3	E3, definition catalog line 133	<code>MaintenanceReinstatementFor</code> (def) <code>lean/SixBirdsFoundationsV/Laws/E3SelfMaintainingReclosure.lean:133</code> <code>Instances.EpisodeLinkedReinstatement</code> (def) <code>lean/SixBirdsFoundationsV/Instances/E3Episode.lean:5</code> <code>Instances.FullyHistoryLinkedReinstatement</code> (def) <code>lean/SixBirdsFoundationsV/Instances/E3Episode.lean:79</code>
Theorem 4.4	E3, instance catalog line 133	<code>Instances.eventTrajectory</code> (theorem) <code>lean/SixBirdsFoundationsV/Instances/E3Executed.lean:20</code> <code>Instances.eventTransition</code> (theorem) <code>lean/SixBirdsFoundationsV/Instances/E3Executed.lean:29</code> <code>Instances.eventHistory</code> (def) <code>lean/SixBirdsFoundationsV/Instances/E3Executed.lean:97</code> <code>Instances.eventApp</code> (def) <code>lean/SixBirdsFoundationsV/Instances/E3Executed.lean:106</code> <code>Instances.eventMaintenance</code> (def) <code>lean/SixBirdsFoundationsV/Instances/E3Executed.lean:116</code> <code>Instances.eventReinstatement</code> (def) <code>lean/SixBirdsFoundationsV/Instances/E3Executed.lean:122</code> <code>Instances.eventE3Reinstatement</code> (theorem) <code>lean/SixBirdsFoundationsV/Instances/E3Executed.lean:151</code> <code>Instances.eventHistoryAgrees</code> (theorem) <code>lean/SixBirdsFoundationsV/Instances/E3Executed.lean:162</code> <code>Instances.eventFullyHistoryLinked</code> (theorem) <code>lean/SixBirdsFoundationsV/Instances/E3Executed.lean:170</code> <code>Instances.eventExecutedReinstatement</code> (theorem) <code>lean/SixBirdsFoundationsV/Instances/E3Executed.lean:187</code> <code>Instances.noEpisodeLinkedWithEmptyHistory</code> (theorem) <code>lean/SixBirdsFoundationsV/Instances/E3Episode.lean:62</code>
Definition 4.6	E2, definition catalog line 109	<code>CarriedInstrumentLevel</code> (structure) <code>lean/SixBirdsFoundationsV/Laws/E2BoundedReflexivity.lean:20</code> <code>CarriedInstrumentLevelOccurrence</code> (def) <code>lean/SixBirdsFoundationsV/Laws/E2BoundedReflexivity.lean:40</code> <code>Instances.DeclaredCapacityMeasure</code> (structure) <code>lean/SixBirdsFoundationsV/Instances/E2Declared.lean:6</code> <code>Instances.DeclaredOccurrence</code> (def) <code>lean/SixBirdsFoundationsV/Instances/E2Declared.lean:43</code> <code>Instances.DeclaredFootprint</code> (def) <code>lean/SixBirdsFoundationsV/Instances/E2Declared.lean:54</code>

Item	Entry and role	Lean declarations
Theorem 4.7	E2, general theorem catalog line 109	Instances.declaredRecordList_nodup (theorem) lean/SixBirdsFoundationsV/Instances/E2Declared.lean:84 Instances.declaredRecordList_subset (theorem) lean/SixBirdsFoundationsV/Instances/E2Declared.lean:104 Instances.declaredRecordList_length (theorem) lean/SixBirdsFoundationsV/Instances/E2Declared.lean:116 Instances.declaredCapacityBound (theorem) lean/SixBirdsFoundationsV/Instances/E2Declared.lean:131 Instances.declaredTowerCapacityBound (theorem) lean/SixBirdsFoundationsV/Instances/E2Declared.lean:148 Instances.copiedRecordReindexNotDeclared (theorem) lean/SixBirdsFoundationsV/Instances/E2Declared.lean:168
Theorem 4.8	E2, instance catalog line 109	Instances.eventCheckedSystem (def) lean/SixBirdsFoundationsV/Instances/E2EventTower.lean:19 Instances.eventCheckedSystemRepair (theorem) lean/SixBirdsFoundationsV/Instances/E2EventTower.lean:39 Instances.eventInspectLower (def) lean/SixBirdsFoundationsV/Instances/E2EventTower.lean:67 Instances.eventLowerInspectionPasses (theorem) lean/SixBirdsFoundationsV/Instances/E2EventTower.lean:70 Instances.eventUpperRecordedCheck (theorem) lean/SixBirdsFoundationsV/Instances/E2EventTower.lean:96 Instances.eventCheckedMeasure (def) lean/SixBirdsFoundationsV/Instances/E2EventTower.lean:128 Instances.eventCheckedTowerAuditsLowerStack (theorem) lean/SixBirdsFoundationsV/Instances/E2EventTower.lean:234 Instances.eventCheckedLevelsDisjoint (theorem) lean/SixBirdsFoundationsV/Instances/E2EventTower.lean:248 Instances.eventCheckedConcreteCapacity (theorem) lean/SixBirdsFoundationsV/Instances/E2EventTower.lean:259 Instances.eventCheckedCapacityBound (theorem) lean/SixBirdsFoundationsV/Instances/E2EventTower.lean:263 Instances.eventCheckedStatusSaturated (theorem) lean/SixBirdsFoundationsV/Instances/E2EventTower.lean:282 Instances.eventCheckedExecutedRepair (theorem) lean/SixBirdsFoundationsV/Instances/E2EventTower.lean:288
Proposition 4.5	E2, imported, weaker form catalog line 109	E2_SelfSoundnessObstruction (theorem) lean/SixBirdsFoundationsV/Laws/E2BoundedReflexivity.lean:831
Proposition A.1	E2, revision note: version-1 measure vacuous catalog line 109	AuditTowerCapacityMeasure (structure) lean/SixBirdsFoundationsV/Laws/E2BoundedReflexivity.lean:267 E2_CapacityBound (theorem) lean/SixBirdsFoundationsV/Laws/E2BoundedReflexivity.lean:1325 Instances.noE2WithOccurringLevel (theorem) lean/SixBirdsFoundationsV/Instances/SpikeModel.lean:104 Instances.spikeNoE2 (theorem) lean/SixBirdsFoundationsV/Instances/SpikeModel.lean:89
Theorem 4.9	E6, general theorem catalog line 205	KKTWitness (structure) lean/SixBirdsFoundationsV/Laws/E6E9PricedAccess.lean:109 GenuineScarcity (def) lean/SixBirdsFoundationsV/Laws/E6E9PricedAccess.lean:41 E6_AttentionKKT (theorem) lean/SixBirdsFoundationsV/Laws/E6E9PricedAccess.lean:265 E6_SlackCollapse (theorem) lean/SixBirdsFoundationsV/Laws/E6E9PricedAccess.lean:372

Item	Entry and role	Lean declarations
Theorem 4.11	E6, general theorem catalog line 205	Instances.weightedCappedObjective (def) lean/SixBirdsFoundationsV/Instances/E6AtCaps.lean:6 Instances.cappedFeasible (def) lean/SixBirdsFoundationsV/Instances/E6Capped.lean:9 Instances.saturatedSlope (def) lean/SixBirdsFoundationsV/Instances/E6AtCaps.lean:9 Instances.weightedSaturatedSupportingPlane (theorem) lean/SixBirdsFoundationsV/Instances/E6AtCaps.lean:13 Instances.CapMultiplierKKT (structure) lean/SixBirdsFoundationsV/Instances/E6AtCaps.lean:45 Instances.capMultiplierKKT_sufficient (theorem) lean/SixBirdsFoundationsV/Instances/E6AtCaps.lean:69 Instances.capExampleKKT (def) lean/SixBirdsFoundationsV/Instances/E6AtCaps.lean:109 Instances.capExampleSlopeZero (theorem) lean/SixBirdsFoundationsV/Instances/E6AtCaps.lean:128 Instances.capExampleGloballyOptimal (theorem) lean/SixBirdsFoundationsV/Instances/E6AtCaps.lean:137 Instances.unitCapExampleGloballyOptimal (theorem) lean/SixBirdsFoundationsV/Instances/E6AtCaps.lean:145
Theorem 4.12	E6, numerical instance catalog line 205	Instances.ActualFamilyForcing (structure) lean/SixBirdsFoundationsV/Instances/E1Periodic.lean:8 GenericallyNovelChallenge (structure) lean/SixBirdsFoundationsV/Laws/E1Internalization.lean:190 StrictSelfExtension (structure) lean/SixBirdsFoundationsV/Laws/E1Internalization.lean:198 InfinitelyManyStrictSelfExtensions (def) lean/SixBirdsFoundationsV/Laws/E1Internalization.lean:202
Definition 4.15	E1, definition catalog line 80	Instances.actualFamilyStrictExtensions (theorem) lean/SixBirdsFoundationsV/Instances/E1Periodic.lean:17
Proposition 4.16	E1, general theorem catalog line 80	Instances.clockedTheory (def)
Theorem 4.17	E1, instance catalog line 80	lean/SixBirdsFoundationsV/Instances/E1Periodic.lean:125 Instances.clockedSystem (def) lean/SixBirdsFoundationsV/Instances/E1Periodic.lean:183 Instances.clockedInstalls (def) lean/SixBirdsFoundationsV/Instances/E1Periodic.lean:204 Instances.clockedOddRepair (theorem) lean/SixBirdsFoundationsV/Instances/E1Periodic.lean:210 Instances.clockedEvenPerturbation (theorem) lean/SixBirdsFoundationsV/Instances/E1Periodic.lean:222 Instances.clockedFamilyEveryEntryInstalled (theorem) lean/SixBirdsFoundationsV/Instances/E1Periodic.lean:343 Instances.clockedUnbounded (theorem) lean/SixBirdsFoundationsV/Instances/E1Periodic.lean:363 Instances.periodicNovel (theorem) lean/SixBirdsFoundationsV/Instances/E1Periodic.lean:286 Instances.periodicStrictAt (theorem) lean/SixBirdsFoundationsV/Instances/E1Periodic.lean:290 Instances.clockedForcing (def) lean/SixBirdsFoundationsV/Instances/E1Periodic.lean:371 Instances.clockedStrictExtensions (theorem) lean/SixBirdsFoundationsV/Instances/E1Periodic.lean:380 Instances.periodicRefinementsDistinct (theorem) lean/SixBirdsFoundationsV/Instances/E1Periodic.lean:296 FiniteForcingStrictnessCertified (structure) lean/SixBirdsFoundationsV/Laws/E1Internalization.lean:235 E1_StrictSelfExtension (theorem) lean/SixBirdsFoundationsV/Laws/E1Internalization.lean:1753 Instances.finiteForcingNoPerpetualNovelty (theorem) lean/SixBirdsFoundationsV/Instances/E1ForcingVacuity.lean:7 Instances.finiteForcingUninhabited (theorem) lean/SixBirdsFoundationsV/Instances/E1ForcingVacuity.lean:40 Instances.E1_StrictSelfExtension_noCertificate (theorem) lean/SixBirdsFoundationsV/Instances/E1ForcingVacuity.lean:64
Proposition A.2	E1, revision note: version-1 certificate uninhabited catalog line 80	

Item	Entry and role	Lean declarations
Definition 4.18	E12, definition catalog line 343	Instances.E12Semantic.PartClosure (structure) lean/SixBirdsFoundationsV/Instances/E12Semantic.lean:13 Instances.E12Semantic.System (structure) lean/SixBirdsFoundationsV/Instances/E12Semantic.lean:20 Instances.E12Semantic.RepairClosed (def) lean/SixBirdsFoundationsV/Instances/E12Semantic.lean:30 Instances.E12Semantic.SelfMaintaining (def) lean/SixBirdsFoundationsV/Instances/E12Semantic.lean:33 Instances.E12Semantic.Candidate (def) lean/SixBirdsFoundationsV/Instances/E12Semantic.lean:37 Instances.E12Semantic.Maximal (def) lean/SixBirdsFoundationsV/Instances/E12Semantic.lean:40 Instances.E12Semantic.Integrated (def) lean/SixBirdsFoundationsV/Instances/E12Semantic.lean:43
Proposition 4.19	E12, general theorem catalog line 343	Instances.E12Semantic.repairClosed_inter (theorem) lean/SixBirdsFoundationsV/Instances/E12Semantic.lean:47 Instances.E12Semantic.integrated_disjoint_or_equal (theorem) lean/SixBirdsFoundationsV/Instances/E12Semantic.lean:67
Theorem 4.20	E12, instance catalog line 343	Instances.E12Semantic.genuineRepairClosure (def) lean/SixBirdsFoundationsV/Instances/E12Semantic.lean:219 Instances.E12Semantic.repairSystem (def) lean/SixBirdsFoundationsV/Instances/E12Semantic.lean:257 Instances.E12Semantic.repairPart_integrated (theorem) lean/SixBirdsFoundationsV/Instances/E12Semantic.lean:288 Instances.E12Semantic.repairedIndividualWitness (theorem) lean/SixBirdsFoundationsV/Instances/E12Semantic.lean:342 Instances.E12Semantic.healthyOnly_closure_loadBearing (theorem) lean/SixBirdsFoundationsV/Instances/E12Semantic.lean:360
Proposition 4.21	E12, negative instance catalog line 343	Instances.E12Semantic.secondSystem (def) lean/SixBirdsFoundationsV/Instances/E12Semantic.lean:145 Instances.E12Semantic.secondLeft_maximal (theorem) lean/SixBirdsFoundationsV/Instances/E12Semantic.lean:161 Instances.E12Semantic.secondRight_maximal (theorem) lean/SixBirdsFoundationsV/Instances/E12Semantic.lean:174 Instances.E12Semantic.second_no_integrated (theorem) lean/SixBirdsFoundationsV/Instances/E12Semantic.lean:187
Definition 4.22	E13, definition catalog line 367	Instances.E13Semantic.System (structure) lean/SixBirdsFoundationsV/Instances/E13Semantic.lean:6 Instances.E13Semantic.Interface (structure) lean/SixBirdsFoundationsV/Instances/E13Semantic.lean:14 Instances.E13Semantic.RepairTransport (structure) lean/SixBirdsFoundationsV/Instances/E13Semantic.lean:24
Theorem 4.23	E13, general theorem catalog line 367	Instances.E13Semantic.compose (def) lean/SixBirdsFoundationsV/Instances/E13Semantic.lean:33 Instances.E13Semantic.compose_payment (theorem) lean/SixBirdsFoundationsV/Instances/E13Semantic.lean:49 Instances.E13Semantic.RepairChain (inductive) lean/SixBirdsFoundationsV/Instances/E13Semantic.lean:151 Instances.E13Semantic.chain_obstruction_bound (theorem) lean/SixBirdsFoundationsV/Instances/E13Semantic.lean:159 Instances.E13Semantic.chain_length_bound (theorem) lean/SixBirdsFoundationsV/Instances/E13Semantic.lean:170

Item	Entry and role	Lean declarations
Theorem 4.24	E13, instance catalog line 367	Instances.E13Semantic.crossTransport (def) lean/SixBirdsFoundationsV/Instances/E13Semantic.lean:185 Instances.E13Semantic.cross_context_zero_to_one (theorem) lean/SixBirdsFoundationsV/Instances/E13Semantic.lean:203 Instances.E13Semantic.cross_context_one_to_zero (theorem) lean/SixBirdsFoundationsV/Instances/E13Semantic.lean:206 Instances.E13Semantic.cross_context_two_step_bound (theorem) lean/SixBirdsFoundationsV/Instances/E13Semantic.lean:216 Instances.E13Semantic.sourceToReceiver (def) lean/SixBirdsFoundationsV/Instances/E13Semantic.lean:73 Instances.E13Semantic.receiverToThird (def) lean/SixBirdsFoundationsV/Instances/E13Semantic.lean:88 Instances.E13Semantic.concrete_composite_payment (theorem) lean/SixBirdsFoundationsV/Instances/E13Semantic.lean:111 Instances.E13Semantic.concrete_composite_reduces (theorem) lean/SixBirdsFoundationsV/Instances/E13Semantic.lean:115
Proposition 4.25	E13, negative instance catalog line 367	Instances.E13Semantic.roleOnlyInterface (def) lean/SixBirdsFoundationsV/Instances/E13Semantic.lean:126 Instances.E13Semantic.roleOnly_mapsRepair (theorem) lean/SixBirdsFoundationsV/Instances/E13Semantic.lean:134 Instances.E13Semantic.roleOnly_no_reduction (theorem) lean/SixBirdsFoundationsV/Instances/E13Semantic.lean:139
Example 4.13	E6, counterexample catalog line 205	argued on paper; not formalized
Remark 4.14	E6, counterexample catalog line 205	argued on paper; not formalized
Proposition 4.10	E6, sufficiency, argued on paper catalog line 205	argued on paper; not formalized
Theorem 5.1	E9, general theorem catalog line 273	E9_CuriosityArgmax (theorem) lean/SixBirdsFoundationsV/Laws/E6E9PricedAccess.lean:479 E6_E9_AccessArbitration (theorem) lean/SixBirdsFoundationsV/Laws/E6E9PricedAccess.lean:587 E6_E9_AccessArbitration_epsilon (theorem) lean/SixBirdsFoundationsV/Laws/E6E9PricedAccess.lean:620 selected_access_move_ratio_bound (theorem) lean/SixBirdsFoundationsV/Laws/E6E9PricedAccess.lean:661 selected_access_move_ratio_bound_epsilon (theorem) lean/SixBirdsFoundationsV/Laws/E6E9PricedAccess.lean:695
Proposition 5.2	E9, general theorem catalog line 273	XiAcqDischarge (def) lean/SixBirdsFoundationsV/Laws/E6E9PricedAccess.lean:406 E9_SameFamilyStrictness (theorem) lean/SixBirdsFoundationsV/Laws/E6E9PricedAccess.lean:414
Proposition 5.3	E15, general theorem catalog line 412	DerivedOfflineBudgetGeometry (structure) lean/SixBirdsFoundationsV/Laws/E15OfflineReclosure.lean:691 kappaOn (def) lean/SixBirdsFoundationsV/Laws/E15OfflineReclosure.lean:722 kappaOff (def) lean/SixBirdsFoundationsV/Laws/E15OfflineReclosure.lean:732 freedAllocation (def) lean/SixBirdsFoundationsV/Laws/E15OfflineReclosure.lean:727 Instances.offlineCapacityFromExhaustion (theorem) lean/SixBirdsFoundationsV/Instances/E15BudgetIdentity.lean:6 Instances.geometryCapacityFromExhaustion (theorem) lean/SixBirdsFoundationsV/Instances/E15BudgetIdentity.lean:29
Proposition 5.4	E4, general theorem catalog line 157	E4_Brittleness (theorem) lean/SixBirdsFoundationsV/Laws/E4RepairCompilation.lean:816
Template S4.1	E1, conditional classification catalog line 80	E1_StatusPartition (theorem) lean/SixBirdsFoundationsV/Laws/E1Internalization.lean:1164 E1_Internalization (theorem) lean/SixBirdsFoundationsV/Laws/E1Internalization.lean:1525 E1_NoGeneratorDichotomy (theorem) lean/SixBirdsFoundationsV/Laws/E1Internalization.lean:1770

Item	Entry and role	Lean declarations
Template 6.1	E3, conditional classification catalog line 133	E3_TwoLevelFixedPoint (theorem) lean/SixBirdsFoundationsV/Laws/E3SelfMaintainingReclosure.lean:2275 E3_StatusPartition (theorem) lean/SixBirdsFoundationsV/Laws/E3SelfMaintainingReclosure.lean:2418 E3_RegressStoppedByE2 (theorem) lean/SixBirdsFoundationsV/Laws/E3SelfMaintainingReclosure.lean:2795 E3_AblationSeparation (theorem) lean/SixBirdsFoundationsV/Laws/E3SelfMaintainingReclosure.lean:2963
Template S4.2	E4, certificate specification catalog line 157	compilationLawful_requiredCoreGatesPass (theorem) lean/SixBirdsFoundationsV/Laws/E4RepairCompilation.lean:190 E4_Compilation (theorem) lean/SixBirdsFoundationsV/Laws/E4RepairCompilation.lean:774 E4_StatusPartition (theorem) lean/SixBirdsFoundationsV/Laws/E4RepairCompilation.lean:843
Definition S4.3	E5, definition catalog line 181	CompleteCollapseRescueInventory (structure) lean/SixBirdsFoundationsV/Laws/E5ReclosureCollapse.lean:756 CollapsedCore (def) lean/SixBirdsFoundationsV/Laws/E5ReclosureCollapse.lean:920
Lemma S4.4	E5, consistency lemma catalog line 181	E5_CollapseFalsifier (theorem) lean/SixBirdsFoundationsV/Laws/E5ReclosureCollapse.lean:2034 E5_ReclosureCollapse (theorem) lean/SixBirdsFoundationsV/Laws/E5ReclosureCollapse.lean:2041 E5_RevivedExcludesLowerPriority (theorem) lean/SixBirdsFoundationsV/Laws/E5ReclosureCollapse.lean:2618 E5_IrreversibleCollapse (theorem) lean/SixBirdsFoundationsV/Laws/E5ReclosureCollapse.lean:2720 E5_SubsidyWithdrawalReclassification (theorem) lean/SixBirdsFoundationsV/Laws/E5ReclosureCollapse.lean:2794
Template S4.5	E7, conditional classification catalog line 229	E7_AlarmTrichotomy (theorem) lean/SixBirdsFoundationsV/Laws/E7Alarm.lean:653 E7_PreemptionSignature (theorem) lean/SixBirdsFoundationsV/Laws/E7Alarm.lean:949
Definition S4.6	E8, definition catalog line 253	ControlPriceComponent (structure) lean/SixBirdsFoundationsV/Laws/E8ControlPrice.lean:153
Lemma S4.7	E8, E8.1, consistency lemma catalog line 253	E8_ControlPrice (theorem) lean/SixBirdsFoundationsV/Laws/E8ControlPrice.lean:1608 E8_ComponentShadowPrice (theorem) lean/SixBirdsFoundationsV/Laws/E8ControlPrice.lean:2007 E8_CompressedSummaryLawfulness (theorem) lean/SixBirdsFoundationsV/Laws/E8ControlPrice.lean:2037 E8_1_AuditCapture (theorem) lean/SixBirdsFoundationsV/Laws/E8ControlPrice.lean:2050 E8_1_CaptureFalsifier (theorem) lean/SixBirdsFoundationsV/Laws/E8ControlPrice.lean:2341 E8_1_MetaAuditBoundedByE2 (theorem) lean/SixBirdsFoundationsV/Laws/E8ControlPrice.lean:2365 BudgetInflationAudit.lineageMustMatchClassifiedMove (theorem) lean/SixBirdsFoundationsV/Laws/E8ControlPrice.lean:637

Item	Entry and role	Lean declarations
Template S4.8	E10–E10.2, conditional classification catalog line 293	E10_CognitiveDemarcation (theorem) lean/SixBirdsFoundationsV/Laws/E10CognitiveDemarcation.lean:1771 E10_ScheduleTrapNull (theorem) lean/SixBirdsFoundationsV/Laws/E10CognitiveDemarcation.lean:2027 E10_ScheduleTrapExcludesLowerPriority (theorem) lean/SixBirdsFoundationsV/Laws/E10CognitiveDemarcation.lean:1963 E10_DecodableCorrelate (theorem) lean/SixBirdsFoundationsV/Laws/E10CognitiveDemarcation.lean:2012 E10_1_Intention (theorem) lean/SixBirdsFoundationsV/Laws/E10CognitiveDemarcation.lean:2042 E10_1_PostHocIntentionFalsifier (theorem) lean/SixBirdsFoundationsV/Laws/E10CognitiveDemarcation.lean:2186 E10_1_PostHocIntentionExcludesLowerPriority (theorem) lean/SixBirdsFoundationsV/Laws/E10CognitiveDemarcation.lean:2155 E10_2_Goal (theorem) lean/SixBirdsFoundationsV/Laws/E10CognitiveDemarcation.lean:2201 E10_2_RewardProxyNotGoal (theorem) lean/SixBirdsFoundationsV/Laws/E10CognitiveDemarcation.lean:2314 E10_2_RewardProxyOnlyExcludesLowerPriority (theorem) lean/SixBirdsFoundationsV/Laws/E10CognitiveDemarcation.lean:2291
Template S4.9	E11, conditional classification catalog line 323	E11_StackActivity (theorem) lean/SixBirdsFoundationsV/Laws/E11InstitutionalRewrite.lean:811 E11_Constitutive (theorem) lean/SixBirdsFoundationsV/Laws/E11InstitutionalRewrite.lean:853 E11_ConditioningOnly (theorem) lean/SixBirdsFoundationsV/Laws/E11InstitutionalRewrite.lean:900 E11_WashoutInert (theorem) lean/SixBirdsFoundationsV/Laws/E11InstitutionalRewrite.lean:941 E11_NCTDObstruction (theorem) lean/SixBirdsFoundationsV/Laws/E11InstitutionalRewrite.lean:966 E11_StatusPartition (theorem) lean/SixBirdsFoundationsV/Laws/E11InstitutionalRewrite.lean:973
Template S4.10	E12, template catalog line 343	E12_Individuation (theorem) lean/SixBirdsFoundationsV/Laws/E12Individuation.lean:711 E12_StatusPartition (theorem) lean/SixBirdsFoundationsV/Laws/E12Individuation.lean:787
Proposition S4.11	E12, template, singleton case catalog line 343	Instances.singletonCandidates (def) lean/SixBirdsFoundationsV/Instances/E12Singleton.lean:5 Instances.E12_singleton_maximal_from_tests (theorem) lean/SixBirdsFoundationsV/Instances/E12Singleton.lean:12 Instances.spikeSelfMaintained (theorem) lean/SixBirdsFoundationsV/Instances/SpikeModel.lean:192
Template 6.2	E13–E13.4, template catalog line 367	E13_Communication (theorem) lean/SixBirdsFoundationsV/Laws/E13RepairTransport.lean:1760 E13_TeachingCapacity (theorem) lean/SixBirdsFoundationsV/Laws/E13RepairTransport.lean:1773 E13_CoercionNull (theorem) lean/SixBirdsFoundationsV/Laws/E13RepairTransport.lean:1786 E13_SymbolicRepair (theorem) lean/SixBirdsFoundationsV/Laws/E13RepairTransport.lean:1799 E13_StatusPartition (theorem) lean/SixBirdsFoundationsV/Laws/E13RepairTransport.lean:1812 E13_CoercionNullExcludesLowerPriority (theorem) lean/SixBirdsFoundationsV/Laws/E13RepairTransport.lean:1992 Instances.transportContextZero (def) lean/SixBirdsFoundationsV/Instances/E13Contexts.lean:5 Instances.transportContextOne (def) lean/SixBirdsFoundationsV/Instances/E13Contexts.lean:10 Instances.E13_two_contexts_distinct (theorem) lean/SixBirdsFoundationsV/Instances/E13Contexts.lean:18 Instances.E13_two_contexts_members (theorem) lean/SixBirdsFoundationsV/Instances/E13Contexts.lean:22

Item	Entry and role	Lean declarations
Template S4.12	E14, conditional classification catalog line 392	E14_ReconsolidationStatus (theorem) lean/SixBirdsFoundationsV/Laws/E14Reconsolidation.lean:1767 E14_LawfulConflictTrichotomy (theorem) lean/SixBirdsFoundationsV/Laws/E14Reconsolidation.lean:2051 E14_OutcomeCollisionExcludesLawful (theorem) lean/SixBirdsFoundationsV/Laws/E14Reconsolidation.lean:2105 E14_RecordRepair (theorem) lean/SixBirdsFoundationsV/Laws/E14Reconsolidation.lean:2217 E14_RecordCoarsening (theorem) lean/SixBirdsFoundationsV/Laws/E14Reconsolidation.lean:2238 E14_StatusedUnresolvedChargedToLambda (theorem) lean/SixBirdsFoundationsV/Laws/E14Reconsolidation.lean:2259 E14_SilentRewriteFalsifier (theorem) lean/SixBirdsFoundationsV/Laws/E14Reconsolidation.lean:2280 E14_RetrievalWithoutTransportControl (theorem) lean/SixBirdsFoundationsV/Laws/E14Reconsolidation.lean:2322 E14_LaunderedProvenanceDefect (theorem) lean/SixBirdsFoundationsV/Laws/E14Reconsolidation.lean:2365
Template S4.13	E15, conditional classification catalog line 412	E15_OfflineReclosureStatus (theorem) lean/SixBirdsFoundationsV/Laws/E15OfflineReclosure.lean:2043 E15_DeficitAlternation (theorem) lean/SixBirdsFoundationsV/Laws/E15OfflineReclosure.lean:2135 E15_OnlineSufficientNoOfflineRequired (theorem) lean/SixBirdsFoundationsV/Laws/E15OfflineReclosure.lean:2182 E15_OfflinePermittedOutsideDeficit (theorem) lean/SixBirdsFoundationsV/Laws/E15OfflineReclosure.lean:2216 E15_SkippedOfflineCascade (theorem) lean/SixBirdsFoundationsV/Laws/E15OfflineReclosure.lean:2241 E15_DecorativeOfflineFalsifier (theorem) lean/SixBirdsFoundationsV/Laws/E15OfflineReclosure.lean:2266 E15_DutyCycleMismatchFalsifier (theorem) lean/SixBirdsFoundationsV/Laws/E15OfflineReclosure.lean:2300 E15_DeficitOnlineFalsifiesNecessity (theorem) lean/SixBirdsFoundationsV/Laws/E15OfflineReclosure.lean:2334 E15_DeficitCounterexampleExcludesLowerPriority (theorem) lean/SixBirdsFoundationsV/Laws/E15OfflineReclosure.lean:2369 E15_OffInventoryFlowCannotCertifyOffline (theorem) lean/SixBirdsFoundationsV/Laws/E15OfflineReclosure.lean:2456

Item	Entry and role	Lean declarations
Template S4.14	E16, conditional classification catalog line 430	E16_AdaptabilityStatus (theorem) lean/SixBirdsFoundationsV/Laws/E16Adaptability.lean:2126 E16_CoherentAdaptability (theorem) lean/SixBirdsFoundationsV/Laws/E16Adaptability.lean:2235 E16_CurrentAuditBlindness (theorem) lean/SixBirdsFoundationsV/Laws/E16Adaptability.lean:2251 E16_RouteManufacturesRepairCapacity (theorem) lean/SixBirdsFoundationsV/Laws/E16Adaptability.lean:2272 E16_ArtifactNotAdaptability (theorem) lean/SixBirdsFoundationsV/Laws/E16Adaptability.lean:2288 E16_FlatNoAdaptability (theorem) lean/SixBirdsFoundationsV/Laws/E16Adaptability.lean:2302 E16_FlattenableNotAdaptability (theorem) lean/SixBirdsFoundationsV/Laws/E16Adaptability.lean:2316 E16_CurrentizableIsLegibleSlack (theorem) lean/SixBirdsFoundationsV/Laws/E16Adaptability.lean:2331 E16_DissipativeNotAdaptability (theorem) lean/SixBirdsFoundationsV/Laws/E16Adaptability.lean:2359 E16_BoundedPerturbationFailureRejected (theorem) lean/SixBirdsFoundationsV/Laws/E16Adaptability.lean:2374 E16_SupportConfoundExcludesLowerPriority (theorem) lean/SixBirdsFoundationsV/Laws/E16Adaptability.lean:2421 E16_LoopAsymmetryAnchor (theorem) lean/SixBirdsFoundationsV/Laws/E16Adaptability.lean:2481 E16_SwappedRouteTrialPopulationRejected (theorem) lean/SixBirdsFoundationsV/Laws/E16Adaptability.lean:457 E16_FlatSecondArmUsesClaimLinkedPair (theorem) lean/SixBirdsFoundationsV/Laws/E16Adaptability.lean:1379 E16_UnrelatedEqualPairCannotReplaceClaimDifference (theorem) lean/SixBirdsFoundationsV/Laws/E16Adaptability.lean:1406

Table S2 gives, for each definition and entry, its name in the theorem catalog and the status under which the main paper states it. The catalog states every entry E1–E16 as a theorem about arbitrary systems; that form is not a proof grade. Where the catalog and the main paper differ, the main paper’s status is the one supported by the Lean statements.

Table S2. Catalog name and status in the main paper.

Entry	Catalog name	Main paper	Line
D1	Repair join	definition; general theorem	54
D2	Predictive surplus, quantified	definition; conditional general theorem	56
D3	Carried record; carried instrument	definition; consistency lemma	60
D4	Endogenous closure system, E-system	definition; consistency lemma	62
D5	Closed-loop scope	definition; consistency lemma	66
D6	Exposure cost; probe economy	definition; consistency lemma	68
E1	Internalization Law	law, weak instance	80
E2	Bounded Reflexivity Law	law, weak instance	109
E3	Self-Maintaining Reclosure Law	law, weak instance	133
E4	Repair Compilation Law	certificate specification	157
E5	Reclosure Collapse Law	definition	181
E6	Attention Law	general theorem	205
E7	Alarm Law	conditional classification	229
E8	Control-Price Law	definition	253
E9	Curiosity Law	general theorem	273

Entry	Catalog name	Main paper	Line
E10	Cognitive Demarcation Law	conditional classification	293
E11	Institutional Rewrite Law	conditional classification	323
E12	Individuation Law	law, weak instance	343
E13	Repair Transport Law	law, weak instance	367
E14	Reconsolidation Law	conditional classification	392
E15	Offline Reclosure Law	conditional classification	412
E16	Adaptability Law	conditional classification	430

S3 Consistency lemmas of the definitions

The following lemmas are proved by projecting the fields of the definitions D3–D6 of the main paper or by contradiction with one of them. Their use is that later certificates cannot silently omit a detector, a gate, a transition, an audit or a source check.

Lemma S3.1 (source exclusion). *A record whose source tag is fallback, independent-pair witness, simulation trace, ablation record, unknown or contradictory is not carried, whatever its flags. A record reached through a step outside the declared scope is not carried. An instrument whose record list is not declared complete is not a carried instrument, even if every listed record is carried.*

Proof. Each clause contradicts one conjunct of Definition 3.6. □

Lemma S3.2 (repair-step consistency). *A lawful repair step entails detection, gate permission, a kernel transition and re-audit; a candidate step lacking any one of the four is not lawful. No repair move has sort P6.*

Lemma S3.3 (closed-loop consistency). *In a closed-loop history every listed repair entry is carried. One listed entry with source tag fallback, unknown or contradictory, or with a false generation or scope flag, defeats closed-loop scope. A fallback entry is not carried even when a different entry with the same payload is.*

Lemma S3.4 (probe-move consistency). *A lawful allocation does not change the support; a lawful acquisition is strict; a lawful retirement removes a probe that was active; and an operator-saturated probe is not a strict acquisition.*

S4 Certificate specifications

This section prints the certificate specifications and definitions of the catalog that the main paper summarizes in Table 4; E3 and E13 are printed in the main paper (Templates 6.1 and 6.2). Each Template gives its data, the certificate that carries the difficult content, and the conclusion that Lean derives from a complete certificate. For a conditional classification the certificate is a complete status record: one carried, branch-valid status record for the claim, and the same status on every genuine record for that claim. After each Template, a *Meaning* sentence says what the conclusion amounts to, and an *Example* gives one status realized in the finite computations of Section S7; the examples are finite checks of the classifiers, not instances of the entries.

S4.1 Endogeny

Template S4.1 (E1: internalization). *Data.* An E -system $\mathbb{S}$, a challenge history H , a recurring challenge class $\mathcal{C}$ with horizon, and a viability readout v with $\Delta(Q_t, F_{\mathcal{C}}, v) \neq \emptyset$ at challenged times. *Certificate.* A complete internalization status record with five statuses: slack, externally subsidized, endogenously repairing, stressed and collapsing. The challenge binds. The set Δ_{endo} is empty; it collects the discharging entries that have a forbidden source, lack generation, scope or carriedness, use an off-kernel move, or omit their audit. The certified inputs listed below are supplied. *Conclusion.* The five statuses are mutually exclusive, and the status is endogenously repairing. The repair family contained in that branch has, at each challenged episode:

- a carried defect and a generated move;
- the installed join $Q_{t+1} = Q_t \vee R_t$;
- a strict reduction $\Delta(Q_{t+1}, F_{\mathcal{C}}, v) \subsetneq \Delta(Q_t, F_{\mathcal{C}}, v)$ on the challenged fibres;
- a committed-state source, a kernel transition, gate passage and re-audit.

Meaning. If the record says the system repairs endogenously and its competitors are ruled out, then every credited repair is a carried join that strictly shrinks the obstruction; the repair family itself is a field of the certificate, not constructed from recurrence. The certified inputs are no-free-distinction [4], promotion-gate soundness [3], a bridge from measured discharge to the absence of residual accrual, and family-level budget feasibility. Perpetual self-extension under a challenge that is always new is the law of Section 4.5. A further statement has the same conditional form: if no repair generator is reachable, a supplied bridge excludes an endogenous family and $\Delta_{\text{endo}} = \emptyset$, then the status is slack, stressed or collapsing, and a binding challenge excludes slack. *Example.* With its repair generator removed, a finite carrier under a split challenge is classified stressed, stressed, collapsing at three successive times, with residuals 1, 2, 3.

Template S4.2 (E4: repair compilation). *Data.* A repair R invoked at least twice for a challenge class $\mathcal{C}$, and a candidate lower operator or support gate (sort $P1$ or $P2$) that would replace the invocation. *Certificate.* A compilation case: the candidate is carried; a certified top-down channel links it to at least two admissible interventions with matched controls; the promotion record is accepted; a memory-only comparator fails to reproduce the descent; the repair is not same-family saturated; an annotation record permits decompilation; in-class descent is verified, $\Delta(Q^{\text{compiled}}, F_{\mathcal{C}}, r) = \emptyset$; the higher invocation is silent; and the higher-priority statuses are excluded. *Conclusion.* The status is compiled; the five statuses exposed, compiling, compiled, mis-compiled and decompiled are mutually exclusive.

Meaning. A repeated repair counts as moved into lower support only when every gate passes; the classification then separates a successful compilation from one that fails outside its class (mis-compiled) or has been undone (decompiled). *Example.* A candidate whose top-down channel is blocked is classified compiling; the same candidate with all gates passing is compiled.

Definition S4.3 (E5: reclosure collapse). Fix a challenge class, a horizon and a complete inventory of typed rescue candidates — carried repairs, boundary updates, apparatus reclosures and probe acquisitions — each with its own move record, budget witness, kernel transition and viability-descent certificate. The system is *collapsed* if no candidate is both admissible within budget and viability-descending. It is *irreversibly collapsed* if moreover the repair generator is unreachable and the apparatus-level viability kernel is empty, both certified for the same system, challenge, horizon and scope.

Lemma S4.4 (collapse consistency). *An admissible in-budget descending candidate refutes collapse. Under a complete status record, the seven statuses viable, stressed, subsidized, suspended, collapsed-recoverable, collapsed-irreversible and revived are mutually exclusive. After a subsidy is withdrawn, a certified absence of active subsidy and the complete record classify the system without presupposing collapse.*

Meaning. Collapse is the absence, in the declared inventory, of any admissible in-budget rescue that restores viability descent; suspension (intact apparatus, operations gated off) and revival (a later rescue with a new apparatus lineage) are separate statuses. The first claim of the lemma holds by definition. *Example.* A carrier with an active kernel but no rescue move is collapsed-recoverable.

S4.2 Priced access and control

Template S4.5 (E7: alarm). Data. *A blind-spot witness z with $z^*(\Xi_C(D \mid L) - \Omega)z > 0$, coupled within the horizon to a failure of a declared viability probe.* Certificate. *A unique total disposition package: one carried, branch-valid disposition record, and the same kind on every genuine record at this witness and horizon.* Conclusion. *Exactly one disposition holds:*

- currentization: *a carried rewrite of the allocator’s constraint set that admits the witness and is charged to a forced-exposure ledger entry;*
- boundary nonclosure: *a failed viability descent with a carried status;*
- lawful discount: *for protocol artifact, currentizable residue, out-of-horizon scope or dissolved coupling.*

A currentization changes the constraint set.

Meaning. To *currentize* a blind-spot direction is to make it part of what the system currently attends to, by changing the constraints of its allocation. A dangerous blind spot is then either currentized in this way, declared an unclosed boundary, or discounted for a stated reason. A move that wins the allocation of Section 4.4 inside the old constraint set is attention, not alarm. A related statement, *alarm fatigue* (E7.1), is argued on paper only: if viability-coupled excess persists and demand exceeds a finite carried preemption capacity, then the channel saturates, a carried threshold repair is charged with the original lineage kept, or the excess stays statused. *Example.* Protocol-artifact witnesses at thresholds 0 and 1 are lawfully discounted.

Definition S4.6 (E8: control price). A carried signal is a *control-price component* for constraint j of a maximization problem with optimal value $\Phi^*(b)$ under constraints of level $b = (b_j)$ when its readout is certified, it is linked to the ledger line of j , its held-out predictions and its stability under budget perturbation pass, and its value matches the multiplier λ_j of a linked multiplier witness when j binds and vanishes when j is slack. A *field* is a complete inventory of components; a compressed summary $a = \mathcal{C}(\lambda_1, \dots, \lambda_k)$ is *lawful* when it predicts admissibility or policy changes beyond the external quotient and collapses when all its components are slack.

Lemma S4.7 (control-price consistency). *A component-lawful case yields its shadow-price component and slack collapse. Summary lawfulness is equivalent to predictive legitimacy together with slack collapse, by definition. Under a complete status record the seven statuses slack-obstructed, proxy-obstructed, summary-redescription, component-lawful, field-lawful, summary-lawful and incomplete-or-unpriced are mutually exclusive.*

Meaning. A control signal counts as a price only when it is linked to the ledger line of its own constraint, matches that constraint’s multiplier, passes its held-out and stability checks, and

vanishes when the constraint is slack. When Φ^* is differentiable and the envelope theorem applies, $\lambda_j = \partial\Phi^*/\partial b_j$ [1, §5.6.3]; this reading is on paper. *Audit capture* (E8.1) is the record in which a budget-setting move raises the carrier's own residual budget, the blind spot against the original lineage stays positive, and the ledger reports health. *Example*. A signal of 1/4 on a slack constraint is slack-obstructed. A budget move from 10 to 15 with the original-lineage blind spot still positive and a healthy report is capture evidence.

Template S4.8 (E10: cognitive demarcation). Data. *A carried package Θ and a challenge class $\mathcal{C}$.* Certificate. *A complete cognitive status record over eight statuses: schedule trap, scaffolding, structural path only, decodable correlate, repair without gate, gate without repair, cognitive, non-cognitive.* Conclusion. *The statuses are mutually exclusive. A cognitive status yields*

$$\text{RepairDescent}(\Theta, \mathcal{C}) \wedge \text{Gate}(\Theta, \mathcal{C}) \wedge \text{Prov}(\Theta, \mathcal{C}), \quad (1)$$

and conversely (1) together with the absence of the six higher-priority kinds of evidence yields the cognitive status.

Meaning. A package with cognitive status repairs, gates action and is the system's own, and no higher-priority evidence is present: *RepairDescent* is a strict reduction of an access or action obstruction by a quotient the package installs; *Gate* is a certified top-down kernel gate for two distinct carried interventions with matched controls and different induced kernel supports; and *Prov* is carried formation and invocation of the package, linked to $\mathcal{C}$. The same form gives *intention* (E10.1: a carried restriction of action support toward a target, strictly before the outcome, certified and coherent under every listed future probe) with five statuses, and *goal* (E10.2: a carried target quotient that persists as the criterion for at least two distinct carried routes) with four. *Example*. A package that an observer can decode but that has no certified difference-making gate is a decodable correlate.

S4.3 Stack, individuation and transport

Template S4.9 (E11: institutional rewrite). Data. *A formed upper package Θ over a lower kernel K_ℓ with closure E_ℓ , and two distinct admissible upper interventions $\Theta_i \neq \Theta_j$ with matched lower controls.* Certificate. *A complete status record over inert, conditioning, stack-active and constitutive, with structural evidence consisting of a certified top-down channel linking the interventions to*

$$\text{supp } K_\ell^{\Theta_i} \neq \text{supp } K_\ell^{\Theta_j} \quad \text{or} \quad E_\ell^{\Theta_i} \neq E_\ell^{\Theta_j}.$$

Conclusion. *The four statuses are mutually exclusive. Stack-active requires structural evidence, an empty stack obstruction and no constitutive evidence; constitutive requires, in addition, a separate compilation certificate. A change of probabilities on unchanged support is conditioning, and a label whose effect washes out is inert.*

Meaning. An upper layer rewrites a lower one only if it changes which lower transitions are possible or how the lower layer completes, not merely how likely they are. *Example*. A weight change from 10^{-3} to 10^{-2} on an allowed transition is conditioning.

Template S4.10 (E12: individuation). Data. *An E-system and a declared list of candidate subcarriers.* Certificate. *For a candidate I : repair, budget and record closure on I ,*

$$\text{Repair}(I) \subseteq I, \quad \text{Budget}(I) \subseteq I, \quad \text{Record}(I) \subseteq I,$$

an interface quotient $B(I)$ certified as the coarsest one through which every declared viability probe descends, and self-maintenance of $B(I)$ by I ; together with maximality among candidates and the absence of a distinct overlapping maximal candidate. A complete status record ranges over integrated, federated, subsidiary, platform-dependent, shadow and non-individuated. Conclusion. The status is integrated; the six statuses are mutually exclusive.

Meaning. An individual is a maximal declared candidate, with no overlapping maximal competitor, whose repairs, budget and records stay inside it and whose coarsest viability-sufficient boundary it maintains itself. *Example.* A candidate whose repairs and records close but whose budget is paid from outside is subsidiary.

Proposition S4.11 (a singleton candidate). *If the candidate list is $[I]$ and I passes the repair, budget, record, viability-boundary and self-maintenance tests, then I is integrated: it is maximal, and no distinct candidate overlaps it.*

Proof. The only candidate is I , so maximality and the absence of a competitor are immediate. $\square$

The semantic form of E12 — repair closure, boundary maintenance, maximality and the exclusion of overlapping rivals, for a finite system — is the law of Section 4.6; the Template adds budget and record closure and a boundary certified as the coarsest viability-sufficient one. *Operational selfhood* (E12.1) is argued on paper only: at the cognitive scale, a *self* is a carried common repair ledger through which the system's role, memory, action and viability probes all descend.

S4.4 Memory and adaptive capacity

Template S4.12 (E14: reconsolidation). Data. A carried record m_t , a retrieval transport $T_c(m_t)$ in context c , and a retrieval conflict: a split pair against the record's own declared claim after the transported record is adjoined. Certificate. Complete inventories of lawful repair, strict coarsening and stasured outcomes, with provenance records for every mutation, and a complete status record over nine statuses: provenance defect, silent rewrite, ordinary read, outcome collision, unrealized disposition, record repaired, record coarsened, stasured unresolved, unstasured conflict. Conclusion. Once the first five statuses and unstasured conflict are excluded, exactly one lawful disposition holds:

repair: m_{t+1} installs the record join of m_t and R_c ;
coarsening: $q^- \preceq q^+$ and $q^+(x) = q^+(x')$, $q^-(x) \neq q^-(x')$ for some x, x' ;
retained debt: $m_{t+1} = m_t$ with a stasured residual charged to Λ_S .

Meaning. When recalling a record exposes a conflict and the five defect statuses and unstasured conflict are excluded, the record is lawfully repaired, made strictly coarser, or kept with the conflict charged as debt; the other statuses record defects in the retrieval or in its bookkeeping. *Example.* The same source record is repaired for one retrieval claim and kept as stasured unresolved debt for another.

Template S4.13 (E15: offline reclosure). Data. Carried closure debt D_t built only from linked stasured E14 residuals, route residues and adequacy residuals; online capacity κ_{on} ; debt accrual a ; and an allocation record as in Proposition 5.3. Certificate. A complete status record over eight statuses: deficit-online counterexample, decorative offline, duty-cycle mismatch, skipped-offline cascade, alternation required, online sufficient, offline optional, offline reclosure rejected. Conclusion. The statuses are mutually exclusive. The alternation-required case yields a strict online deficit $a > \kappa_{\text{on}}$, recurrent offline phases that gate exchange to zero and assign the freed allocation to internal discharge with linked debt discharge in each phase, and an observed duty cycle within a tolerance declared before the horizon.

Meaning. A system that accrues debt faster than it can discharge online is certified to need alternating offline phases only when the phases, the freed budget and the duty cycle are all on record; Proposition 5.3 gives the capacity of such a phase. *Example.* At debt accrual 5 the claim is alternation required; at accruals $7/2$ and 4 it is online sufficient.

Template S4.14 (E16: adaptability). *Data.* Two admissible routes γ, η on one comparison support that end at the same current practice quotient; a held-out challenge family; complete endpoint-conditioned trial populations and their repair-capacity distributions. *Certificate.* A complete status record over eight statuses: support confound, artifact, flat, flattenable, currentizable slack, dissipative, coherent adaptability, adaptability rejected; with the controls for artifact, flattening, currentization, continuation, support confounding and bounded perturbation. *Conclusion.* The statuses are mutually exclusive. Coherent-adaptability evidence yields that the two endpoints are equivalent for every current event, and two claim-linked probability records for the same challenge class with different probabilities.

Meaning. Two histories that look the same now are recorded as leaving different future repair capacity, after every declared control has failed to explain the difference away; the difference is read from the record, not identified as an effect of the route. *Example.* Two routes to current-event-equivalent endpoints give repair probabilities $3/4$ and $1/4$ for the same challenge; six control regimes realize one status each, among them coherent adaptability.

S5 Field lists

This section lists, entry by entry, the fields of the records that the main paper names: the certificates of the Templates, the status branches of each conditional classification, and the data of the definitions. A field is a hypothesis of the corresponding Lean statement; nothing listed here is derived from the type of an E-system. Throughout, a *claim* is the full tuple that a status record is about (system, challenge class, horizon, and the records the entry names), and every field is scoped to that claim: evidence for one claim cannot be used for another.

S5.1 The definitional layer

Carried record (D3). A declared coordinate map $\rho_{\text{of}} : Z \rightarrow R$; a declared trajectory τ with a legitimate start; an index $n_0 \geq n_{\text{start}}$ such that every step from n_{start} to n_0 is in scope and lies in $\text{supp } K$; the value $\rho = \rho_{\text{of}}(\tau(n_0))$; a source tag among committed state and audited cell records; generation and scope flags equal to true. The source tags are committed state, audited cell records, independent-pair witness, simulation trace, ablation record, fallback, unknown and contradictory.

Carried instrument (D3). An instrument of the audit calculus of Foundations III; a predicate saying which instrument records are carried; lists of visibility, threshold and check-rule records, each carried; and a proposition declaring the lists complete.

E-system (D4). A theory package with carrier, kernel support, legitimate-start predicate, declared trajectory, scope predicate, lens, definability structure, completion, audit, and a formedness proposition with its proof; carried-record policies for defect, move and audit records; an active carried instrument with the predicates *detects*, *gate allows* and *re-audits*; a carried ledger (a policy, a list of entries, a completeness proposition with its proof, carriedness of every entry); a repair generator sending each defect to a move with a sort in P1–P5, a payload, a carried move record and a budget line; and an admissibility predicate.

Lawful repair step (D4). Defect carried; detected at z ; gate allows the generated move at z ; move admissible at z ; budget line in the ledger; $(z, z') \in \text{supp } K$; re-audit at z' with the audit record; audit record carried.

Challenge history and closed-loop scope (D5). Episodes, each with a time, a challenge class, a source quotient, a declared family, a target readout and an optional obstruction witness; repair-typed entries (defect, move or audit), each with an entry class carrying an identifier, a source tag and generation and scope flags; an association of entries to episodes, declared for every listed entry; a completeness proposition. Closed-loop scope is completeness together with carriedness of every listed entry.

Probe economy (D6). A finite catalog; a carried active family with a support list, a weight function and an optional operator family; a carried exposure budget and spend in the ledger; a saturation predicate; allocation, acquisition and retirement moves, each with pre- and post-family carriedness records, and for acquisition strictness, catalog membership, budget admissibility and absence from the prior support.

S5.2 Endogeny

E1. The endogenous branch contains, for each challenged episode: the defect record and its generated move; pre- and post-states; the audit record; the repair payload and the installed refinement with $Q_{t+1} = Q_t \vee R_t$; a strict challenged-fibre reduction of the split-pair obstruction; a committed-state source; kernel realization; gate passage; and re-audit. The other branches are: slack (the challenge is not binding); externally subsidized (a discharging entry that is not endogenous and no valid endogenous family); stressed (viability preserved while a statused residual accrues); collapsing (a viability probe fails). The certified inputs are no-free-distinction, finite-forcing strictness, promotion soundness, the bridge from measured discharge to absence of residual accrual, and family-level budget feasibility.

E2 (first version). A carried instrument level has a level index, a tag, an instrument, its own instrument record, an audit predicate on lists of lower indices, a lower-stack target, and the ledger entries and audit records it uses. It occurs if its own record is carried by its instrument and by the system instrument, its instrument is carried, its ledger entries are in the ledger, its audit records are carried, and every target index is below its own. The first-version measure has injective, disjoint tags for the three record sorts; a record universe per state with size equal to the capacity; for every occurring level a record set of exactly its counted records, inside every universe, with positive size; and disjointness between occurring levels of different index.

E2 (declared measure). As in Definition 4.6 of the main paper: depth, a duplicate-free slot list covering every index below the depth, a level at each slot with matching index and a proof that the level occurs, the tag data, the universes and capacities, the slot record sets with the counting, inclusion, disjointness and positivity fields. A declared tower lists the level of each slot at its index.

E2 (forced stratification). An audited-repair trigger; the audit-stack record of Foundations III with finite records and an admissible lower stack; a bridge from the tower to that stack; non-saturation; and a fresh-level bridge requiring every carried level at the target index to carry a tag above every tag below it and to audit exactly the lower stack.

E3. Apparatus records at t and $t + 1$ with their carried records (time, gate instrument, threshold records, boundary, audit data, apparatus record, used ledger entries and audit records); a maintenance operator with its record, its action on states, and its ledger entries and audit records; a reinstatement record (time, source and target states, operator, pre-, post- and output apparatus records, ledger entry) that is carried with an allowed source, whose fields match, whose post apparatus is the operator applied to the source, whose ledger entry is in the ledger and whose source and target are joined by a kernel move. The maintained branch adds an object fixed point, object record coherence, a maintained step (distance at most ε without challenge; a linked E1 repair with defect, move, join installation, kernel transition and re-audit under challenge), an empty maintenance obstruction, no subsidy, and regress-stop evidence. The maintenance obstruction is nonempty if a required component (instrument, gates, thresholds, boundary, audit data) is absent, has a forbidden source, fails generation or scope, is off-kernel, or has a valid reinstatement omitted from the audit.

E4. A candidate window of at least two carried invocations of the same attributed repair; an obstruction-reduction predicate and an idempotence-stability predicate, independent of each other; a compiled operator record of sort P1 or P2, indexed by the challenge class and the repair refinement; the lawfulness gates (certified top-down channel, accepted promotion record, failed memory-only comparator, non-saturation, annotation record); verified in-class descent; silence of the higher invocation. Statuses in priority order: decompiled (a linked decompilation event), mis-compiled (the compiled core with an unstatused out-of-class obstruction), compiled, compiling (an attributed candidate missing a gate, descent or silence), exposed (no attributed candidate).

E5. Rescue candidates of four types (repair, boundary update, apparatus reclosure, probe acquisition), each with a move record, a budget witness, a kernel transition and a descent certificate for the declared viability probes; a list containing every candidate. Irreversibility inputs: unreachable repair generator and empty apparatus-level viability kernel, coherently scoped. Statuses in priority order: revived (a prior subsidized or collapsed status, a later carried rescue, a new apparatus lineage), subsidized (an active external carrier with its own scope, budget and descending move, and no internal falsifier), suspended (intact apparatus, operations gated off), viable, stressed (additionally, scoped residual accrual), collapsed-irreversible and collapsed-recoverable (the latter with its own recoverability evidence).

S5.3 Priced access and control

E6. A budget record: spend, budget, their ledger entries (in the ledger and certified by the economy) and budget admissibility of the move. A multiplier witness: $\lambda \geq 0$, $\mu_p \geq 0$, the two slackness equations and stationarity on the support. A binding budget is $\text{spend} = \text{budget}$ with $\text{spend} \leq \text{budget}$; a slack budget is $\text{spend} < \text{budget}$.

E7. Currentization: a carried constraint rewrite linking old and new constraint sets, a forced-exposure ledger entry naming the witness, the new set different from the old and admitting the witness, and passing no-overread and null-mode checks. Boundary nonclosure: a failed viability descent and a carried status. Lawful discount: a carried reason among protocol artifact, currentizable residue, out of horizon, dissolved coupling. The package requires one branch-valid carried record and the same kind for every carried record at the witness and horizon.

E8. A certified signal readout; a declared constraint record; a ledger-alignment comparator; held-out prediction data; a dual-stability record under declared budget perturbations; a linked multiplier

witness. A binding component: the constraint binds, the multiplier is the declared dual price, alignment, held-out prediction and stability pass. A slack component: multiplier and price component vanish, or a carried residual explains the deviation. Capture evidence: a linked budget rise by the classified move, a positive blind spot against the original accepted lineage, and a healthy ledger report; the audit statuses are capture-claim-rejected, capture-detected, residual-discharge-verified, meta-audit-capacity-blocked and meta-audit-missing.

E9. An admissible acquisition: catalog membership, absence from the support, strictness, a budget record for the acquisition move with spend at most budget, and the viability-risk predicate. A complete candidate list contains every admissible acquisition. An access move is an allocation increment on the current support or a strict acquisition, priced by a common ratio.

E10. Repair descent: a nonempty obstruction before, a quotient installed by the package, and a strict reduction after. Gate: two distinct carried interventions, matched controls, a channel record linked to that comparison, accepted gate status, and different induced kernel supports. Provenance: carried formation and invocation records, the invocation linked to the class. Statuses in priority order: schedule trap (an exogenous schedule explains the support change and a complete gate inventory has no accepted gate), scaffolding (no carried package provenance), structural path only (a lower-support difference and a structural path with the channel blocked), decodable correlate, repair without gate, gate without repair, cognitive, non-cognitive. Intention: provenance, a top-down restriction strictly before the outcome, strict inclusion of the restricted action support, and coherence under every member of a nonempty carried list of future probes; statuses post-hoc, approximate support, target-incoherent, intention, not intention. Goal: provenance, top-down orientation, a carried target quotient, and evaluation of every listed route among at least two distinct carried routes; statuses reward proxy only, route unstable, goal, not goal.

S5.4 Stack, individuation and transport

E11. Two carried intervention records and a matched-control record; a top-down channel of Foundations III referring to that comparison, accepted and passing its effect gate; a support witness (z, z') with different support indicators, or different closures. The stack obstruction is inhabited by an independently carried parameter-only comparison (parameters differ, support and closure unchanged) with the same outcome. Statuses in priority order: constitutive (structural effect, empty stack obstruction, a carried compiled-rewrite record with a certified bridge), stack-active, conditioning (parameter-only comparison and no structural effect), inert (washout, blocked channel, or no evidence).

E12. Repair closure: some repair is attributed to I and every attributed repair has its target in I . Record closure: some genuine activity is attributed to I and every entry it generates closes on I . Budget closure: a feasible exposure budget attributed through the carried trajectory, every such entry closing on I . Boundary: a quotient certified sufficient for the declared viability probes that is a function of every admitted sufficient quotient (so it is the coarsest), and self-maintained by a boundary apparatus reinstated in the sense of E3 with source state in I . Maximality ranges over the declared candidate list. Statuses: integrated; federated (a non-individuating candidate properly inside a maximal one, or two overlapping maximal candidates); subsidiary (repair and records close, budget does not); platform-dependent (all closures and a sufficient coarsest boundary, not self-maintained); shadow (repair and budget close, records do not); non-individuated.

E13. A paid, carried bridge between the systems; a carried lawful receiver move; a strict, installed reduction of the receiver’s obstruction in the shared class; interface mediation; repair-role preservation. Teaching adds a later probe with the source absent, the bridge closed and the repair carried by the receiver from committed state. Coercion null: a comparator-certified fully forced interaction, a carried, complete and empty free-response census, no receiver-committed transported repair, and equal before and after obstructions. Symbolic repair: a paid carried bridge; a declared context family with at least two entries; reactivation in every listed context; stability of the repair role; same-family strictness. Statuses in priority order: coercion null, symbolic, taught, communicated, scaffolded or primed, influence only, transport rejected.

S5.5 Memory and adaptive capacity

E14. A nonempty duplicate-free retrieval-context family; a complete transport inventory, single-valued per context; context dependence (two contexts whose transports differ); the conflict, a split pair after the transported record is joined to the current quotient; formed carried source records. Repair: a lawful repair step, a carried next-version record, exact retrieval and context linkage, honest provenance, installation of the record join. Coarsening: the before quotient refines the after quotient, a strictness pair, and a carried audit that the lost distinction is unsupported. Retained debt: unchanged record and a positive carried residual charged to an existing ledger entry. Statuses in priority order: provenance defect, silent rewrite, ordinary read, outcome collision, unrealized disposition, record repaired, record coarsened, statused unresolved, unstatused conflict.

E15. Closure debt from canonically linked E14 statused residuals, certified route residues and adequacy residuals. The allocation record: carried, linked to the scope, declared before the horizon, a binding budget with carried entries, a positive online external share, a nonnegative online internal share, online exhaustion, offline exchange zero, the offline internal share equal to the online internal plus external shares, offline exhaustion. Alternation required: strict online deficit, recurrent lawful offline phases on the declared flow inventory with linked debt discharge, a duty-cycle prediction, a tolerance declared before the horizon and the observed duty cycle within it. Statuses in priority order: deficit-online counterexample, decorative offline, duty-cycle mismatch, skipped-offline cascade, alternation required, online sufficient, offline optional, rejected.

E16. A carried route-transport package with the comparison base, the transport core of the holonomy development, two routes and their endpoints; current equivalence of the endpoints and a predictive witness separating them; complete endpoint-conditioned trial inventories with eligibility, success, protocol, measurement time and binding budget common to both arms and positive denominators; a held-out challenge with $p_\gamma(C) \neq p_\eta(C)$ for the claim-linked pair; an accepted reconciliation with any E15 route residue on the same package and pair; and four freedom certificates: no eligible honest counterpart changes the comparison base; no admissible completion equalizes the residue; no same-base refinement makes it current; no own continuation erases it and every declared in-bound perturbation preserves it. Statuses in priority order: support confound, artifact, flat, flattenable, currentizable slack, dissipative, coherent adaptability, adaptability rejected.

S6 Non-implications

Table S3 lists, for each entry, statements that the results do not support, with the separating example or the missing hypothesis. A *separating example* is a record that satisfies the stated hypotheses and

fails the tempting conclusion; the examples marked *finite* are realized in the finite computations of Section S7, and the others are stated here. Table S4 lists the exclusion statements checked in Lean: each says that, for one claim, raw evidence of a higher-priority kind excludes a set of lower judgments.

Table S3. Non-implications and separating examples.

Entry	Does not follow	Separating example or reason
D1	The join is admissible as an access object.	The product construction is unrestricted; admissibility of joint access is a separate condition of Foundations IV.
D1	The join reduces an obstruction.	$q \vee q$ has the fibres of q ; nothing new is separated.
D2	A finer quotient has smaller closure deficit.	Refinement can worsen the score; the comparison is a hypothesis.
D2	Scores at different horizons are comparable.	They refer to different futures; a bridge is needed.
D3	A committed-state record with a false generation flag is carried.	The flag is a conjunct.
D3	A fallback copy of a carried record is carried.	Provenance attaches to the entry, not the payload value.
D3	Every listed record carried makes the instrument carried.	Without the completeness declaration it is not.
D4	A lawful step shows recurrence or maintenance.	The step is one lawful transition.
D4	A successful readout after the fact replaces a missing detection, gate, transition or audit.	Each is a separate conjunct (Lemma S3.2).
D5	A closed-loop history contains a repair.	The empty history is closed-loop.
D6	Turning a dormant catalog probe from weight zero to positive weight is allocation.	It changes the support and is an acquisition with its own obligations.
E1	Every persistent system has a carried repair source.	Persistence without discharge can be slack or stressed; the status record decides.
E1	External discharge is self-repair.	An externally supplied repair that reduces the obstruction makes Δ_{endo} nonempty (finite).
E1	Every discharge is strict novelty.	Strictness needs the genericity certificate; a same-family relabel discharges nothing (finite).
E2	Every finite E-system has a capacity-bounded self-model.	The first-version measure exists for no system with an instrument level (Proposition A.1); the bound needs a declared measure.
E2	An audited repair forces a new level.	The next level is a certified input.
E2	The bound gives an optimal tower depth.	Only with a cost and error model; in the finite example the optimum depth 2 is a property of the declared loss.
E3	Object persistence shows maintained closure.	A crystal-grade control persists with no maintenance operator (finite).
E3	A reinstatement shows a repair episode.	The first-version condition holds with an empty history.
E3	The ablation separation predicts the curves.	The curves are measured inputs.

Entry	Does not follow	Separating example or reason
E4	Repetition is compilation.	A window that reduces the obstruction but drifts, and one that is stable but does no work, both fail (finite).
E4	Compilation is always brittle.	Brittleness needs an exhibited out-of-class split pair.
E5	Activity is viability.	A carrier with an active kernel and no rescue move is collapsed-recoverable (finite).
E5	Over-budget descent refutes collapse.	The falsifier must be admissible within budget (finite).
E5	Subsidy withdrawal causes collapse.	The post-withdrawal status is classified afresh; viability is possible (finite).
E6	The multiplier witness identifies an optimum.	Example 4.13; and, for a linear objective, Remark 4.14, where the supplied marginals are not the objective's derivatives.
E6	A binding budget gives a positive price.	A flat objective spends the budget at price zero.
E6	Budget spend equals the objective.	The two are separate data; the numerical optimum at a cap supplies no carried spend record and no link between spend and objective.
E7	A large residual triggers an alarm.	Viability coupling is a separate predicate; uncoupled protocol artifacts are discounted (finite).
E7	Winning the attention allocation is an alarm.	An alarm changes the constraint set.
E7	A carrier must alarm.	The package supplies the disposition.
E8	Every control signal equals a multiplier.	Only a component is claimed.
E8	A budget rise is capture.	A rise with a falling residual is discharge (finite).
E8	A report flag about one move shows capture by another.	Lineage must match the classified move.
E9	The maximizer is the system's choice.	The policy needs its own argmax certificate.
E9	Acquisition installs a repair.	Only a later record linking it to an obstruction does.
E10	A decodable state is cognitive.	Decodable correlate status.
E10	Repair and gate evidence together give cognitive status.	Structural-path evidence may coexist with an accepted gate; the higher-priority negations are needed.
E10	A restriction recorded after the outcome is an intention.	Post-hoc status.
E10	Reward correlation is a goal.	Reward-proxy status.
E11	A probability change is a rewrite.	Conditioning status on unchanged support.
E11	A structural downward path is an accepted channel.	The imported blocked-channel witness.
E11	A compiled repair is constitutive.	No bridge from E4 to the constitutive status is proved.
E12	A membrane is a boundary.	A candidate may close all three flows while a declared viability probe fails to descend through the membrane quotient.
E12	A shared database is a self.	The probes of all four domains must descend through one carried ledger.
E12	Budget closure can be dropped.	A subsidiary closes repair and records but not budget (finite).
E13	Force and an empty census give coercion null.	Absence of committed receiver repair and an unchanged obstruction are separate.

Entry	Does not follow	Separating example or reason
E13	A context family of length two has two contexts.	The family type allows a repeated entry.
E13	Behaviour change is teaching.	Teaching needs source-absent receiver-carried repair.
E14	Mutation under retrieval is an error.	Repair and coarsening are lawful outcomes.
E14	A changed record is a silent rewrite.	Only with no lawful outcome, no disposition and empty mutation provenance.
E14	A disposition tag realizes an outcome.	Unrealized-disposition status.
E15	Growing debt makes offline phases necessary.	Necessity is a field of the certificate.
E15	$a \leq \kappa_{\text{on}}$ gives online sufficiency.	The full online-sufficient case is required.
E15	A cascade shows a skipped phase.	The cascade records an all-online deficit only.
E16	Unequal probabilities show that a route creates capacity.	No intervention bridge is proved.
E16	Equality for another pair shows flatness.	Only the claim-linked pair counts.
E16	A protocol effect is adaptability.	The control battery must be passed.

Table S4. Exclusion statements checked in Lean.

Entry	Higher-priority evidence	Excluded for the same claim
E5	revived	subsidized, suspended, viable, stressed, both collapsed statuses
E10	schedule trap	scaffolding, structural path only, decodable correlate, repair without gate, gate without repair, cognitive, non-cognitive
E10	decodable correlate; schedule trap	cognitive
E10.1	post-hoc restriction	approximate support, target-incoherent, intention, not intention
E10.2	reward proxy only	route unstable, goal, not goal
E13	coercion null	symbolic, taught, communicated, scaffolded or primed, influence only, transport rejected
E14	outcome collision	record repaired, record coarsened, stasured unresolved
E14	silent rewrite; ordinary read	the three lawful outcomes
E15	deficit-online counterexample	decorative offline, duty-cycle mismatch, skipped cascade, alternation required, online sufficient, offline optional, rejected
E15	decorative offline; duty-cycle mismatch	alternation required
E15	a flow record outside the inventory	genuine and decorative offline phases with that record
E16	support confound	artifact, flat, flattenable, currentizable slack, dissipative, coherent adaptability, rejected
E16	artifact; flat; flattenable; currentizable slack; dissipative	coherent adaptability

Each exclusion is proved from explicit negations of higher-priority evidence in the lower branches together with uniqueness of the status for one claim. None of them derives the completeness of an inventory, the honesty of a comparator, a source classification or a causal identification.

S6.1 Further tempting inferences

Table S5. Further tempting inferences and what blocks them.

Tempting inference	Why it fails
Refining the quotient raises the predictive surplus.	Refinement can worsen a closure score; the surplus inequalities need their comparison and chain-rule hypotheses (Proposition 3.5).
The device produced its own history.	The history and its associations are declared with the device; Theorem 4.4 checks that they agree with the executed step.
The link in Theorem 4.4 carries information about which defect was repaired.	The defect sort of the device has one element; a richer defect model is needed (Section 7.4).
Every finite E-system has a bounded, non-final self-model.	The capacity bound needs a declared measure (Theorem 4.7).
An audited repair forces a new audit level.	The order of the level tags and the next level's audit relation are a certified input (Section 4.3).
E2 gives an optimal audit depth.	Only with a cost and error model.
The budget spend equals the value of the allocation.	They are separate data; the multiplier witness of Theorem 4.9 names neither an objective nor a feasible set.
The individual of Theorem 4.20 is found from the dynamics alone.	Maximality is relative to the declared boundary regions (Definition 4.18).
The argmax of Theorem 5.1 is what the system chooses.	The theorem constructs a maximizer; a policy's choice is shown optimal by its own argmax certificate.
A strict acquisition installs a repair.	Only a later record linking it to an obstruction does.
Offline phases are necessary whenever debt grows faster than online capacity.	Necessity is part of the E15 certificate; the scalar deficit alone does not give it, and $a \leq \kappa_{\text{on}}$ alone does not give online sufficiency (Template S4.13).
A skipped-offline cascade shows that a scheduled phase was skipped.	It records an all-online deficit with growing debt, a timed alarm and later stress (Template S4.13).
Unequal repair probabilities after two routes show that a route creates capacity.	The E16 record reports the difference; a causal reading needs an intervention bridge (Template S4.14).
The internalization record constructs the repair family.	The family is a field of the endogenous branch (Template S4.1).
A compiled repair is brittle on every other class.	Proposition 5.4 certifies only an exhibited split pair.
The alarm Template shows that a carrier must alarm.	The disposition is supplied by the package (Template S4.5).
A decodable internal state is cognitive.	A readable state without certified difference-making is a decodable correlate (Template S4.8).
Repair, gate and provenance evidence together give the cognitive status.	The six higher-priority kinds of evidence must also be absent (Template S4.8).
A signal that tracks a constraint is its shadow price.	Linkage, held-out prediction, stability and slack collapse are required (Definition S4.6).

Tempting inference	Why it fails
A long run without collapse shows viability.	Collapse is relative to the declared rescue inventory and viability probes (Definition S4.3).
A structural downward path is an accepted channel.	The blocked-channel witness imported from Foundations III separates them (Template S4.9).
A compiled repair is constitutive.	No theorem maps an E4 compilation to the constitutive status of E11.
Transport across two contexts gives symbolic repair.	Symbolic repair asks that a token reactivate a carried repair in every context of a declared family; Theorem 4.24 transports a repair relation, not a token (Template 6.2).
Force and an empty census give a coercion null.	The absence of committed receiver repair and an unchanged obstruction are separate conditions (Template 6.2).
A changed record after retrieval is a silent rewrite.	Only with no lawful outcome, no typed disposition and no mutation provenance (Template S4.12).

S7 Finite computations

S7.1 The finite carriers

Repair-World is a family of finite controlled Markov carriers whose states have the form

$$z = (y, q, L, r, \Lambda, A),$$

with a public state y , a carried quotient table q , an active probe family L , a record ledger r , a budget vector Λ and one or more carried audit levels A . The environment supplies a finite challenge process with controllable recurrence, drift, novelty and viability coupling. The common kernel implements the carrier, source, quotient, probe, ledger and audit interfaces; E4, E14, E15 and E16 use their own finite checkers over the same vocabulary, on rings of 18, 20 and 30 states, a probe shop and a memory wheel. The computations use exact rational arithmetic. For each entry the expected outputs — statuses, counts, rational values and the controls that must be rejected — were fixed before the checker was run, and a comparison passes only on equality with the fixed value.

The computations show that the hypotheses of each certificate can be met on a finite carrier, that every status of each classification has a constructed case, and that the constructed false positives are rejected. They do not show that any catalog of probes or rescues is empirically complete, that any domain reading is an instance, or that the frequencies of statuses generalize.

Table S6. Finite comparisons by entry; all pass.

Entry	Comparisons	What is compared
E1	15	repair source, generator ablation, five statuses, subsidy, saturation, schedule, audit omission
E2	13	tower extension, capacity, saturation, linkage, status, loss optimum
E3	17	two-level closure, regress stop, ablation curves, source and audit defects
E4	16	compilation, gate and operator variants, brittleness, decompilation, drift, blocked candidates
E5	20	seven statuses, four rescue types, subsidy, withdrawal, reachability, inventory, budget

Entry	Comparisons	What is compared
E6, E9	14	multipliers, slack, finite differences, saturation, acquisition, arbitration, proxy, salience
E7	12	excess, latency, preemption, survival, lawful discount, false and missed alarms, linkage
E8	21	component, field and summary statuses, slack, proxy, lineage, capture, capacity
E10	26	eight cognitive, five intention and four goal statuses, linkage and timing controls
E11	20	support and closure rewrite, conditioning, washout, channel block, constitutive record, scoping
E12	19	six statuses, three closures, boundary sufficiency, coarsestness, maximality, overlap
E13	19	communication, teaching, coercion null, symbolic repair, scaffolding, class, role, bridge
E14	21	nine statuses, retrieval trigger, three outcomes, collision, provenance, context, coarsening
E15	46	eight statuses, debt sources, budget geometry, duty cycle, cascade, inventory, gating, priority, timing
E16	69	eight statuses, route distributions, control inventories, reconciliation with E15, probabilities, perturbations
Total	348	

The capacities and footprints in the E2, E3 and E8 computations are record counts computed directly on the finite carriers. They are not instances of either Lean capacity measure.

S7.2 Values by entry

E1. The challenge is at baseline at times 0–1 and splits at 2–4. The initial obstruction has one pair and the installed join removes it. The carrier with its repair generator spends 2 units of a budget of 3 and is endogenously repairing at times 2–4. With the generator removed the statuses are stressed, stressed, collapsing, with residuals 1, 2, 3. External subsidy and a schedule trap both remove the obstruction but leave one endogeny defect and are classified externally subsidized; a same-family relabel leaves the obstruction and is stressed.

E2. On a carrier of capacity 8, towers of depth 0–4 have footprints 0, 2, 5, 8, 10, so depths up to 3 fit. Triggered audit rotates depth $1 \rightarrow 2 \rightarrow 3$; the saturated trigger stays at 3. The fresh levels have tags 20 and 30 and targets $\{0\}$ and $\{0, 1\}$; a level with the wrong next index is rejected. With loss equal to audit cost plus repair error, the losses at depths 0–3 are $18/24, 14/24, 13/24, 14/24$, so depth 2 is optimal for this loss.

E3. Without challenge the apparatus distance is $1/8$ under tolerance $1/4$. After the maintenance channel is cut, the apparatus distance is $0, 1/8, 3/8, 1/2, 5/8, 3/4$ and the object distance $0, 1/16, 1/16, 1/8, 3/16, 5/16$; they separate first at time 2, while the maintained reference stays at $1/8$. A crystal-grade control never separates. Absent-record, fallback, off-kernel, out-of-scope, uncarried and audit-omission controls each make the maintenance obstruction nonempty.

E4. Two candidates, an operator rewrite and a support gate, pass all gates and are compiled. Adding an out-of-class witness makes one mis-compiled; adding a decompilation event makes one

decompiled. A blocked channel, a rejected promotion, a memory-only reproduction, same-family saturation and unverified descent each give compiling. A one-invocation window and a P6 compiled sort are rejected at construction. A window that reduces the obstruction while its payload drifts fails stability, and one that is stable without reducing the obstruction fails reduction.

E5. An 18-state ring realizes all seven statuses; repair, boundary update, acquisition and apparatus reclosure each supply a separate viable rescue. External rescue gives subsidized while the internal inventory is still collapsed. A carrier with an active kernel and no rescue move is collapsed-recoverable. An over-budget descending move is rejected as a falsifier. After withdrawal of a subsidy the scenario is classified viable once absence of active subsidy is checked.

E6 and E9. In the probe shop the exact optima at budgets 1, 2, 3 are $(4/5, 1/5)$, $(7/5, 3/5)$ and $(2, 1)$ with equal selected-probe ratios $25/9$, $25/16$ and 1; budget 4 is slack at $(2, 1)$ with price zero. Discharge rises $14/3 \rightarrow 27/4 \rightarrow 8 \rightarrow 8$. A proxy allocation discharges 2 at every budget and at budget 3 leaves residual trace $45/4$ against $21/4$. The largest excess direction is probe A with magnitude 3 and moves to probe B with magnitude 8 after the declared change of factor. Finite differences at perturbation $1/20$ give selected ratios $100/37$, $8/3$ at budget 1, $75/49$, $50/33$ at 2 and $60/59$, $40/39$ at 3, with spreads that shrink along perturbations $1/5$, $1/10$, $1/20$. Strict candidates C, AC and BC each discharge $1/4$, with ratios $1/4$, $1/8$, $1/16$; same-family candidates discharge nothing. Joint arbitration spends budgets 1–3 on allocation and acquires C only at budget 4, raising total discharge from 8 to $33/4$.

E7. Residual and excess are 0, 1, 4, 9 for hazard levels 0–3. Monitoring budgets 1, 2, 3 give detection latencies 2, 1, 0, and budget 0 never currentizes. In three hazards, preemption selects the alarm action and survives three times, while the old ratio ranks the ordinary action at 1 above the alarm action at $1/2$ and fails three times. Protocol artifacts at thresholds 0 and 1 are discounted. Boundary nonclosure and alarm fatigue are not exercised.

E8. The binding component spends its unit budget with multiplier $3/2$; a slack component spends $1/2$ with multiplier 0; a slack signal of $1/4$ is obstructed. A proxy with multiplier $4/5$, signal $9/10$ and drift $1/5 > 1/20$ fails alignment and stability. Genuine discharge lowers the residual from 5 to 2; the capture case raises the budget from 10 to 15 with a positive original-lineage blind spot and a healthy report; a rise from 10 to 20 with the residual falling from 9 to 4 is not capture.

E10. All eight cognitive, five intention and four goal statuses are realized. Controls reject a support comparison linked to the wrong intervention, a repair credited to the wrong class, a post-outcome restriction, an approximate support shift, an incoherent future probe, a reward-only explanation, an omitted route evaluation, and uncarried intention or goal records.

E11. A support rewrite and a closure-only rewrite are realized, with all four statuses. A weight change from 10^{-3} to 10^{-2} is conditioning. A support difference with a blocked channel is inert. The constitutive status appears only with the compiled-rewrite record.

E12. On a 30-state ring: integrated candidates; a subsidiary with closure signature (repair, budget, record) = (T, F, T) ; a platform-dependent candidate (T, T, T) whose boundary is not self-maintained; a shadow (T, T, F) ; federated fragments and overlapping maximal candidates. A candidate with all three closures and no sufficient boundary is non-individuated, and a smaller closing candidate

is rejected when a declared larger one closes. A two-class boundary is sufficient and coarsest; a three-class one is sufficient but not coarsest.

E13. Clean communication, teaching with the source absent, coercion null, stable symbolic reactivation, scaffolding, priming and different-class influence are realized. Eight failure controls — unpaid bridge, uncarried receiver repair, role change, context instability, role drift, saturation relabel, a parameter-only channel, a flow crossing with no bridge — are transport rejected.

E14. Under a loose memory budget the outcome counts (repair, coarsen, unresolved) are $(6, 1, 1)$; under a tight one $(1, 6, 1)$. This interaction with the budget is observed, not derived. One source record is repaired for one retrieval claim and unresolved for another. A duplicate context or two transports for one context reject the setup; wrong-direction and non-strict coarsening fail.

E15. A lawful alternation has observed and predicted duty cycle $1/4$; a mismatch has $1/3$ against $1/4$, a gap of $1/12 > 1/24$. The debt from three sources sums to 20; the budget identity is $6 = 4 + 2$, and fitted gains $11/2$ and $13/2$ are rejected. At accrual rates $7/2$ and 4 the claim is online-sufficient, and at 5 alternation is required.

E16. In the memory-wheel carrier the two endpoints are current-event equivalent and $p_\gamma = 3/4$, $p_\eta = 1/4$ for the repair challenge. The current loop action is trivial and the predictive action is not; 256 continuations are enumerated. One case of each of flat, artifact, flattenable, currentizable, dissipative and coherent is realized. A perturbation of $3/4$ lies outside the declared bound $1/2$; an in-bound perturbation of $1/4$ destroys coherence and gives rejection. Stored probabilities that disagree with the counted ratio are rejected.

S8 Life-facing synthetic data and a biological protocol

The carrier-side form of E1–E16 makes living organization a primary domain of application, but substrate independence does not by itself turn a biological analogy into a theorem instance. This section therefore keeps three evidential levels separate. First, two computational substrates supply finite evidence, with matched controls, for life-like maintenance, drive separation, route dependence, and controlled higher-level readout [5]. Second, those measurements are compared with the recognition conditions of the E-laws, with every missing bridge stated explicitly. Third, a concrete planarian regeneration study is specified as an auditable biological protocol. That protocol has not yet been run and is not reported as evidence for any theorem. The numbers below are those reported in the life-focused development [5]; no result of the main paper depends on them.

S8.1 Two finite synthetic substrates

The life-focused study instantiates two finite computational substrates rather than treating “life” as a decorative interpretation [5]. The particle substrate has positions on a two-dimensional torus, writable bonds w_{ij} , bounded counters a_i and n_i , a finite packaging field S , and optional operator-token and meta-layer fields. Its principal lenses expose the packaging field, operator/interface mismatch, and windowed currents, affinities, motifs, and protocol observables. The neural/meta-layer substrate is a layered finite lattice with bounded site variables, auxiliary fields, within-layer tokens W under a global budget, and inter-layer tokens K under a per-site budget. Its lenses

Material	Evidence status	Admissible use here
Particle and neural substrates	Synthetic experiments with specified criteria and matched controls	Finite evidence for the reported operational signals; not evidence that the substrates are alive and not, without a bridge record, instances of an E-law
E-law correspondence below	Interpretive recognition map	Identifies which measured fields could populate an E-law bridge and which typed hypotheses remain unverified
HoloLife-1 planarian protocol	Proposed and unrun	Preregistered biological target, positive criteria, and falsifiers; no positive biological result is claimed

Table S7. Evidence-status discipline for the life-facing material.

expose stroboscopic currents, cross-layer mismatch, motif inventories, dictionaries, and intervention-conditioned decoding.

In both substrates, packaging is organized by the already specified dynamics-induced completion template

$$E_{\tau,f}(\nu) = Q_f((U_f\nu)P^\tau).$$

The experiments do not compute the associated idempotence defect. Their stability claims are therefore operational maintenance and retention claims, not measurements of exact closure. Some neural “snapshot” observables also depend on an entropy-production ledger; these are properly lenses on an extended diagnostic state rather than unqualified observables of the physical lattice state.

For the particle substrate, accepted $+1$ and -1 updates of each writable coordinate type $y \in \{w, a, n, S\}$ give

$$J_y = \frac{N_y^+ - N_y^-}{T_w}, \quad A_y = \log \frac{N_y^+ + 1}{N_y^- + 1}, \quad \Sigma_{\text{mem}} = \sum_y J_y A_y.$$

The reported entropy-production quantities in both substrates are likewise accepted-move proxies. They omit rejected moves and, under the documented conventions, proposal-ratio terms. They are not exact path-space Kullback–Leibler audits, need not be nonnegative in finite windows, and are used only through null calibration and matched-regime separation.

Primitive	Particle realization	Neural/meta-layer realization
P1	writable bonds w_{ij}	local W - and K -token exchange kernels
P2	bounded apparatus counters a_i	conserved token budgets and economy diffusion
P3	explicit protocol phase/clock	deterministic kernel cycle keyed to step index
P4	quantized counters n_i	template-flip kernel in the later/meta presets
P5	writable packaging field S	barrier/closure field and structured token-mediated packaging
P6	drive toggle and μ -contexts	nonconservative work term with a separate drive toggle

Table S8. Realizations of the primitive roles in the two synthetic substrates. A primitive being implemented does not establish an E-law whose setup uses that primitive.

S8.1.1 Null calibration and drive separation

The audit-first sequence begins with P3 and P6 disabled. The relevant proxy audits are near zero while the dynamics remain nondegenerate (table S9). Only after this calibration is P6 enabled under a matched regime. The resulting separation is reported in table S10. These are measurements of a controlled drive channel; they do not establish biochemical metabolism, thermodynamic autonomy, or an exact arrow-of-time functional.

Substrate	Null regime	Exact reported proxy summary
Particles	P3 off, P6 off	$\Sigma_{\text{mem}} = \text{sigmaMem} \approx 3 \times 10^{-4}$; M6 motif affinities = 0; accepted-move EP proxy $\approx 3.5 \times 10^{-6}$ with confidence-interval half-width $\approx 4.7 \times 10^{-5}$, containing zero
Neural	P3 off, P6 off	Specified bound $ \text{epMicroRateWindowLast} \leq 2 \times 10^{-4}$ for the specified seeds; at 24×24 over three seeds, mean EP $\approx -3.9 \times 10^{-6}$, maximum confidence-interval half-width $\approx 2.8 \times 10^{-4}$, and acceptance ≈ 0.54

Table S9. Executed null-regime calibration. The sign of the finite-window neural proxy is not assigned thermodynamic meaning.

Substrate	Regime	Exact reported separation
Particles	Null, P6 off	$\text{sigmaMem} \approx 3 \times 10^{-4}$; M6 affinities = 0; EP proxy $\approx 3.5 \times 10^{-6}$ with a confidence interval containing zero
Particles	Drive, P6 on	$\text{sigmaMem} \approx 2.1 \times 10^{-3}$; M6W ≈ 0.814 , M6N ≈ 0.374 , M6A ≈ 0.392 ; EP proxy $\approx 4.26 \times 10^{-3}$
Neural	Null, P6 off	$ \text{null_mean} \leq 5 \times 10^{-4}$ with nondegenerate acceptance
Neural	Drive, P6 on	$\text{drive_mean} \geq 10^{-3}$ and $\text{drive_mean} - \text{null_mean} \geq 5 \times 10^{-4}$ at $B_k = 2$ and $\eta_{\text{drive}} = 2$

Table S10. Executed P6 null-drive separation with P3 held off.

S8.1.2 Protocol effect without a directionality overread

The neural P3 experiment compares a deterministic kernel schedule with a matched control at one observation lens. The schedule changes the stroboscopic-current statistic in every reported row (table S11), but the protocol is keyed to the step index. Unless protocol phase is included in the state, the observed process is time-inhomogeneous. Accordingly, the measurement is evidence of a route/protocol effect and is not used as a directionality certificate.

Setting	Control L2	Protocol L2	Relative change	Accept (control)	Accept (protocol)
seed 1, $\eta = 0.5$ (additional)	0.0349	0.0262	0.250	—	—
seed 1, $\eta = 1.0$ (additional)	0.0379	0.0271	0.284	—	—
seed 1, $\eta = 2.0$ (specified)	0.0311	0.0255	0.179	—	—
seed 1, $\eta = 1.0$ (replication)	0.0379	0.0271	0.284	0.0103	0.0107
seed 2, $\eta = 1.0$ (replication)	0.0326	0.0280	0.141	0.0116	0.0107
seed 3, $\eta = 1.0$ (replication)	0.0313	0.0280	0.105	0.0107	0.0108

Table S11. Executed neural protocol diagnostic. L2 is the reported `strobe_current_l2_window`. The $\eta = 2$ row is the specified case; the other rows are additional settings.

S8.1.3 Maintenance and viability probes

The maintenance experiments ask whether a finite system preserves or recovers a structured target under noise, traversal, hazard, or deadline. The targets, hazards, and deadlines are imposed by the experimental design. Thus the results establish controlled maintenance competencies, not a free metabolism and not an autonomous origin of the repair objective.

Substrate	Task	Reported outcome
Particles	Code maintenance	Null: $\text{err} = 0.43 \pm 0.09$, $\text{Sdiff} = 6.86 \pm 0.25$, $\text{epRate} = 0$. Drive: $\text{err} = 0$, $\text{Sdiff} = 0.83 \pm 0.06$, $\text{epRate} = 0.016$
Particles	Gated traversal	Ungated: $\text{err} = 0.23$, $\text{Sdiff} = 0.49$. Gated moving: $\text{err} = 0$, $\text{Sdiff} = 0$. Gated static: $\text{Sdiff} = 2.58$ and no recovery
Particles	Deadline selection	$\Delta\text{uptimeTail} = 0.285$ with confidence interval $[0.14, 0.43]$; clock cost $\Delta\text{EPClockRate} = 0.083$
Neural	Paired hazard response	$\text{spike_paired} = 0.063$, $\text{realloc_paired} = 0.015$, $\text{recovery_paired} = 0.060$; declared reallocation threshold 0.002

Table S12. Executed maintenance and viability probes. The neural hazard response is measured with paired runs.

S8.1.4 Refined-lens motifs and controlled readout

The neural substrate also exposes motif histograms and motif-transition statistics at a refined diagnostic lens. The inventory experiment requires coverage at least 0.60, Jensen–Shannon divergence at least 0.01, and propagation score at least 0.02; all three seeds exceed those thresholds. The transition experiment reports the transition divergence in the same finite regime.

Experiment	Seed	Coverage (pre)	Coverage (hazard)	JSD (pre/hazard)	JSD (transition)	Propagation
inventory	1	1.0	1.0	0.0312	–	0.644
inventory	2	1.0	1.0	0.0324	–	0.587
inventory	3	1.0	1.0	0.0316	–	0.609
transition	1	1.0	–	0.0312	0.0383	0.644
transition	2	1.0	–	0.0324	0.0418	0.587
transition	3	1.0	–	0.0316	0.0424	0.609

Table S13. Executed motif-inventory and motif-transition results.

Four further experiments inject controlled labels and evaluate the resulting motif-level statistics against a within-run shifted-label null. Here “semantics” means intervention-conditioned discriminability and recovery of injected labels, not linguistic meaning, awareness, or subjective content. The p -values in table S14 are experiment-local shift-null controls; no global family-wise error claim is made.

Three other readout experiments failed their shifted-null criteria. They count against the readout claims, not for them. In particular, the successful intervention-conditioned decoding does not follow merely from the presence of motifs.

S8.2 What the synthetic evidence does and does not instantiate

The measurements above give life-facing substance to the abstract catalog, but they are not silently promoted to E-law instances. The recognition map in table S15 states the strongest source-faithful relation.

No result above directly instantiates E4 compilation, E11 support rewrite, E12 individuation, or E13 repair transport. The two substrates do, however, show why living organization belongs

Experiment	Seed	Metric	Value	p -value	Reported separation
directional focus	1	focus delta	1.478	0.005	spike control 0.269
directional focus	2	focus delta	1.453	0.000	spike control 0.225
directional focus	3	focus delta	1.521	0.000	spike control 0.384
dictionary polarity	1	dictionary delta	0.064	0.005	JSD in/out 0.032
dictionary polarity	2	dictionary delta	0.108	0.005	JSD in/out 0.060
dictionary polarity	3	dictionary delta	0.075	0.005	JSD in/out 0.039
phrase alignment	1	alignment	0.999	0.005	JSD out/in 0.665
phrase alignment	2	alignment	1.000	0.005	JSD out/in 0.671
phrase alignment	3	alignment	1.000	0.005	JSD out/in 0.667
phrase decoding	1	token accuracy	1.000	0.005	JSD out/in 0.665
phrase decoding	2	token accuracy	1.000	0.005	JSD out/in 0.671
phrase decoding	3	token accuracy	1.000	0.005	JSD out/in 0.667

Table S14. Executed intervention-conditioned readout and shifted-label-null results.

E-law surface	Existing life-facing signal	Missing bridge or control
E1 Internalization and E3 Self-Maintaining Reclosure	Driven code maintenance, gated traversal, perturbation recovery, and explicit costs are finite repair/maintenance candidates	The target and hazard are externally installed. A full instance still requires a carried defect record, carrier-produced source repair, the E1 closed-loop record, and the E3 apparatus reclosure and audit chain
E5 Reclosure Collapse	Null, ungated, and gated-static failures provide genuine separating controls for maintenance claims	Failure of an experimental task is not yet the typed exhaustion of moves, budget, and reclosure machinery required by E5
E6 Priced Exposure and E9 Strict Probe Acquisition	The neural carrier has explicit W/K budgets; the particle carrier exposes finite diagnostic families	No complete native/probe catalogue, strict-family test, Ξ residual, positive-cost ledger, or KKT allocation certificate links these runs to E6 or E9
E7 Alarm and E8 Control-Price	The paired neural hazard produces a spike, resource reallocation, and recovery; the particle deadline run reports a benefit and an explicit proxy cost	The E7 disposition inventory and carried alarm record are not built, and the E8 signal is not linked to a declared binding constraint, dual variable, lineage audit, and capture controls
E10 Cognitive Demarcation	Motifs, dictionaries, and intervention-conditioned decoding furnish decodable correlates under controls	Decodability is not cognition. A positive E10 instance additionally needs the same carrier’s repair-and-gate evidence, carried decision link, counterfactual ablations, and status partition
E14 Reconsolidation and E15 Offline Reclosure	The controlled representations provide candidate records to examine	No retrieval-contingent record revision, canonical E14 status, on-line/offline debt flow, exchange census, or duty-cycle theorem is measured here
E16 Adaptability	The P3 study establishes a repeatable protocol-dependent change in one stroboscopic statistic	A protocol effect alone is not coherent adaptability. E16 still requires fixed support, matched current endpoints, exact endpoint-conditioned repair distributions, E15 reconciliation, complete counterpart inventories, and bounded-perturbation survival

Table S15. Cautious recognition map from executed synthetic evidence to the E-law surfaces. Every row records both positive material and the missing bridge.

beside cognition and society rather than underneath them: maintenance, resource allocation, route dependence, and controlled readout are already nontrivial carrier-side phenomena before any cognitive interpretation is attempted.

S8.3 HoloLife-1: a proposed biological instantiation

The synthetic evidence does not substitute for a biological test. The HoloLife-1 proposal stated here targets planarian regeneration, motivated by work on bioelectric control and regenerative pattern memory [2]. Its central object is not a black-box predictor and not a claim that the organism “believes” a target morphology. It is a finite, audit-led descent/repair claim: two histories that agree at a declared current quotient may separate under a future recut challenge, and the proposed study asks whether a carried bioelectric predicate and route record lawfully repair that failure.

Finite carrier and audit ledger. Let $H = \{h_i\}$ be a frozen finite set of regeneration-history records. Each record contains

$h =$ (source/study, species, fragment type,
amputation geometry, treatment route, washout state,
time/stage, current anatomy, current molecular markers, bioelectric readout,
recut challenge, future outcome, reset intervention, audit ledger).

The audit ledger is carrier data. It records instruments, declared thresholds, visible and suppressed fields, empirical bridges, species and protocol compatibility, remaining residuals, and any target coarsening. In particular, public molecular atlases may populate a declared bridge for the current molecular quotient only when species, time, and protocol compatibility are recorded; they may not be silently merged with the perturbation histories.

Four quotients and one held-out challenge. The proposed quotients are

$$Q_0 = Q_A \vee Q_G, \quad Q_B = Q_0 \vee h_B, \quad Q_R = Q_B \vee \rho, \quad M = Q_\Theta.$$

Here Q_A records current visible anatomy, Q_G the declared current molecular/marker state, h_B a finite bioelectric predicate, ρ a route-memory residue, and Q_Θ the future target-morphology quotient. The held-out map $F_{\text{recut}} : H_t \rightarrow H_{t+\Delta}$ is read into

$$r_\Theta F_{\text{recut}}(h) \in \{W, C, D, R, A\},$$

where W is a wild-type single-headed target, C a cryptic altered target, D a double-headed target, R a reset/repaired target, and A an abnormal or failed package. These labels are proposed finite outcome classes; they do not earn P5 objecthood merely by being assigned.

Test 1: split-pair obstruction. The primary target is the already specified descent obstruction

$$\Omega(Q_0; F, r_\Theta) = \{(h, h') : Q_0(h) = Q_0(h') \wedge r_\Theta F(h) \neq r_\Theta F(h')\}.$$

The positive first-stage criterion is $\Omega(Q_0; F, r_\Theta) \neq \emptyset$. This says that the declared present anatomy/marker quotient is inadequate for the future recut readout. It is stronger and more falsifiable than reporting predictive accuracy alone.

Test 2: strict bioelectric source repair. The candidate h_B must first split at least one old fiber:

$$h_B \notin \Sigma_{Q_0}, \quad I(h_B \mid Q_0) = \frac{1}{N} \sum_{C \in H/Q_0} \min(n_{C,0}, n_{C,1}) > 0.$$

It must then reduce the preregistered obstruction,

$$\hat{\omega}(Q_B) \ll \hat{\omega}(Q_0), \quad Q_B = Q_0 \vee h_B,$$

rather than merely improve a fitted score. Entropy-matched random predicates are explicit comparators. Target coarsening is recorded as a different repair: collapsing head/tail outcomes to “viable” cannot be credited as an explanation of the original distinction.

Test 3: hiddenness under exposure. The HoloLife-1 proposal uses the empirical witness-pair frequency below. To avoid identifying it with D2’s closure-deficit surplus without a bridge, write it here as

$$S_{\text{pair}}(Q, M) = \Pr[h \sim_Q h' \wedge h \not\sim_M h'],$$

with positive pattern

$$S_{\text{pair}}(Q_0, M) > 0, \quad S_{\text{pair}}(Q_B, M) \ll S_{\text{pair}}(Q_0, M).$$

Threshold and treatment-strength sweeps test whether the reduction is tied to exposure of the determining predicate rather than to an arbitrary large effect. A residual that remains after exposure stays stashed rather than being declared solved. Reading it as an instance of D2 would require the registered closure-deficit score or an explicit bridge from this finite pair frequency to that score.

Test 4: route memory without the protocol trap. The comparison fixes two routes:

γ : untreated regeneration $\rightarrow$ current single-headed state $\rightarrow$ recut,

η : transient bioelectric perturbation $\rightarrow$ washout $\rightarrow$ current single-headed state $\rightarrow$ recut.

The proposed holonomy-memory signature is

$$Q_0(\gamma h) = Q_0(\eta h), \quad M(\gamma h) \neq M(\eta h),$$

with the route residue $\rho_{\gamma,\eta}$ added on the source side through $Q_R = Q_B \vee \rho_{\gamma,\eta}$. This would be a physiological formed record, not a cognitive state. It would not by itself establish P6 or thermodynamic directionality. Any P6 claim must separately lift treatment, phase, and protocol into state and survive the corresponding audit.

Test 5: package formation. The proposed target states earn P5 status only through the existing finite completion test

$$E_{\tau,q}(\mu) = U_q(Q_q(\mu K^\tau)), \quad \delta_{\tau,q} = \frac{1}{2} \max_z \sum_{z'} \left| (E_{\tau,q}^2 - E_{\tau,q})(z, z') \right|.$$

The preregistered positive condition is $\delta_{\tau,Q_\Theta} \ll 1$ together with at least two nontrivial stable packages. A constant map, unstable classifier, or post hoc outcome bin does not pass.

Test 6: lower-layer gating or rewrite. For target package θ , the proposed lower-layer kernel is

$$K_B^\theta(b, b') = \Pr(B_{t+\Delta} = b' \mid B_t = b, \Theta_t = \theta, \phi_t),$$

where ϕ_t includes phase, treatment, washout, recut protocol, and instrument state. The registered comparisons are kernel divergence, support symmetric difference, cycle-rank change, and spectral-gap change. A support or feasibility change is a P2 candidate; a change in reachable classes, absorbing states, cycle rank, spectral behavior, or the induced packaging map is a P1 candidate. Small parameter drift alone is below the registered threshold. The distinctive positive target is a difference among K_B^W, K_B^C, K_B^D under the same lifted external protocol.

Test 7: residual and currency discharge. Let $L = \{Q_A, Q_G\}$ be the native current probes and let $D = \{h_B, \rho_{\gamma, \eta}, Q_\Theta\}$ be the proposed dissolving family. The existing adequacy residual is

$$\Xi_C(D \mid L) = K_{DD} - K_{DL} K_{LL}^\dagger K_{LD}.$$

The protocol requires the residual after adding the bioelectric and route records to fall below a declared budget:

$$\Xi_C(D_{\text{remaining}} \mid L \vee h_B \vee \rho) < \text{declared budget}.$$

If it does not, the biological needle remains undissolved. A later shadow price may be interpreted only when the constrained budget problem, binding lineage, and dual certificate are all actually supplied; a fitted coefficient is not an E8 control price.

Role-status ledger. The proposal ends in the finite status table of table S16, not in an untyped claim that all six primitives are present.

Role	Positive witness sought	Demotion or failure
P1 rewrite	K_B^θ changes reachable classes, absorbing states, cycle rank, spectral gap, or package map by target state	Differences vanish after phase, protocol, treatment, and instrument lifting
P2 gating	Target package changes the support or feasibility of bioelectric/regenerative transitions	Only weak parameter drift; no support or feasibility change
P3 holonomy	Routes agree at Q_0 but differ at M	Difference is explained by an unrecorded treatment, time, or protocol field
P4 staging/forking	W, C, D, R, A form a typed branch tree with transition records and stage indices	Labels are post hoc outcome bins
P5 packaging	Q_Θ has low idempotence defect and multiple stable packages	Constant map or unstable classifier
P6 audit	Thresholds, instruments, bridges, residuals, repairs, coarsenings, and nonclaims are carried and stasured	Any silent bridge, silent translation, or unstasured residual

Table S16. Proposed HoloLife-1 role-status ledger. Every row is a test to be run, not an observed biological result.

S8.3.1 Preregistered positive package

A strong positive result requires the conjunction in table S17. Passing only one or two rows would support a narrower biological statement, not the proposed full-stack conclusion.

S8.3.2 Preregistered falsifiers

The protocol is designed to make failure informative and easy to report.

Preregistered criterion	Licensed conclusion
$\Omega(Q_0; F, r_\Theta) \neq \emptyset$	The declared present anatomy/marker quotient fails descent to future form
$h_B \notin \Sigma_{Q_0}$ and $\hat{\omega}(Q_B) \ll \hat{\omega}(Q_0)$	The bioelectric predicate is a strict source extension that repairs, rather than merely redescribes, the obstruction
$S_{\text{pair}}(Q_B, M) \ll S_{\text{pair}}(Q_0, M)$	Predictive hiddenness is reduced under the declared exposure
$Q_0(\gamma h) = Q_0(\eta h)$ but $M(\gamma h) \neq M(\eta h)$	The declared routes require a source-side memory residue
$\delta_{\tau, Q_\Theta} \ll 1$ with at least two nontrivial stable packages	Target morphology passes the proposed P5 package test
Target-conditioned K_B^θ differ under the same lifted protocol	The formed package is associated with lower-layer gating or rewrite, as classified by the registered support and kernel tests
$\Xi_C(D_{\text{remaining}} \mid L \vee h_B \vee \rho)$ is below the declared budget	The residual is discharged in the registered currency rather than hidden by prose
All role channels are statused	No primitive is credited through an unrecorded bridge or residual

Table S17. Conjunctive positive criteria for the proposed biological protocol.

Observed failure	Consequence
Q_0 already descends to future target morphology	There is no split-pair obstruction and no hidden target package relative to the declared current quotient
h_B is definable from Q_0	There is no strict bioelectric source extension
An entropy-matched random predicate repairs descent as well as h_B	The proposed source repair lacks the registered biological specificity
Only target coarsening repairs the obstruction	The distinction has been erased rather than explained
Route memory disappears after treatment, time, phase, and protocol are lifted	The apparent holonomy was a protocol or omitted-variable artifact
δ_{τ, Q_Θ} remains high or only a constant package survives	Target morphology has not earned P5 objecthood
The adequacy residual remains above budget after h_B and ρ are added	The proposed needle remains unresolved
K_B^θ does not vary with target package under the fixed lifted protocol	This evidence pack does not support the lower-layer rewrite hypothesis
Any bridge or residual cannot be statused	The full Six-Birds biological package is rejected even if individual correlations remain

Table S18. Falsifiers for the proposed HoloLife-1 study.

Implementation boundary. The proposal calls for a frozen evidence pack containing perturbation route, current morphology, future recut outcome, reset behavior, and declared bioelectric readouts; digitized voltage-map features; and molecular atlases used only through explicit compatibility bridges. The core analysis is a finite obstruction/repair/audit certificate using finite-state estimation, uncertainty intervals, bootstrap or permutation controls, and entropy-matched random predicates. A simulator may be attached only as a model-realization map. It cannot replace the primary biological carrier or convert this unrun protocol into present evidence.

Scope of a future positive result. Even the full conjunction would not define biological life, prove that all living systems realize E1–E16, establish consciousness, or derive a thermodynamic arrow from protocol holonomy. It would provide a finite biological realization of a narrower normal form: current-quotient failure, strict physiological source repair, exposure-sensitive predictive surplus, route-carried memory, package formation, lower-layer transition change, and audited residual discharge. Until the protocol is executed, that sentence states the target of the experiment and nothing more.

S9 Formal status

S9.1 The library

The Lean 4 library has six modules for the definitions D1–D6, fifteen for the entries E1–E16 (E6 and E9 share one) and twenty for the instances. It uses the vendored Lean interfaces of Foundations III, of the adequacy-residual development and of the holonomy development, imports no other external library, contains no `sorry`, and declares no axioms of its own. The first-version capacity measure of E2 and the statements that use it are part of the library; the main paper states which of them are vacuous.

S9.2 Axioms

Table S19 gives, for every theorem named in Table S1, the axioms reported by `#print axioms`. The record for the definition and entry modules is dated 2 September 2026 and covers 187 theorems of those modules; 98 of them use no axiom. The record for the instance modules is the transcript of 26 September 2026. Only the three standard axioms of Lean’s kernel occur.

Table S19. Axioms of the theorems in the concordance.

Declaration	Axioms	Record
<code>join_refines_left</code>	none	2 Sep 2026
<code>join_refines_right</code>	none	2 Sep 2026
<code>vee_greatest_lower_bound</code>	none	2 Sep 2026
<code>join_idem</code>	none	2 Sep 2026
<code>join_comm</code>	none	2 Sep 2026
<code>join_assoc</code>	none	2 Sep 2026
<code>join_well_defined</code>	none	2 Sep 2026
<code>predictiveSurplusValue_nonnegative_iff</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>	2 Sep 2026
<code>predictiveSurplusValue_identical</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>	2 Sep 2026
<code>predictiveSurplusValue_exact_baseline</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>	2 Sep 2026
<code>predictiveSurplusXi_nonnegative_of_psd</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>	2 Sep 2026
<code>predictiveSurplusXi_chain_rule_monotone</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>	2 Sep 2026

Declaration	Axioms	Record
predictiveSurplusXi_same_family_saturation	propext, Classical.choice, Quot.sound	2 Sep 2026
not_carriedSource_fallback	none	2 Sep 2026
not_carriedSource_independent_pair_witness	none	2 Sep 2026
not_carriedSource_simulation_trace	none	2 Sep 2026
not_carriedSource_ablation_record	none	2 Sep 2026
not_carriedSource_unknown	none	2 Sep 2026
not_carriedSource_contradictory	none	2 Sep 2026
not_carriedRecord_of_step_not_in_scope	none	2 Sep 2026
not_carriedInstrument_of_incomplete_inventory	none	2 Sep 2026
lawfulRepairStep_requires_detection	none	2 Sep 2026
lawfulRepairStep_requires_gate	none	2 Sep 2026
lawfulRepairStep_requires_kernel_transition	none	2 Sep 2026
lawfulRepairStep_requires_reaudit	none	2 Sep 2026
not_lawfulRepairStep_without_detection	none	2 Sep 2026
not_lawfulRepairStep_without_gate	none	2 Sep 2026
not_lawfulRepairStep_without_kernel_transition	none	2 Sep 2026
not_lawfulRepairStep_without_reaudit	none	2 Sep 2026
no_repairSort_p6	none	2 Sep 2026
closedLoopScope_complete	none	2 Sep 2026
closedLoopScope_entry_carried	none	2 Sep 2026
not_closedLoopScope_of_listed_source_fallback	none	2 Sep 2026
not_closedLoopScope_of_listed_source_unknown	none	2 Sep 2026
not_closedLoopScope_of_listed_source_contradictory	none	2 Sep 2026
not_closedLoopScope_of_listed_generatedByS_false	none	2 Sep 2026
not_closedLoopScope_of_listed_inScope_false	none	2 Sep 2026
fallback_entry_cannot_borrow_carriedness	none	2 Sep 2026
lawfulAllocation_no_new_active_probe	none	2 Sep 2026
lawfulAcquisition_requires_acquisitionStrict	none	2 Sep 2026
lawfulRetirement_requires_active	none	2 Sep 2026
lawfulRetirement_probe_absent_after	none	2 Sep 2026
not_lawfulAcquisition_of_sameFamilySaturated	none	2 Sep 2026
xi_sameFamilySaturated_not_acquisitionStrict	propext, Classical.choice, Quot.sound	2 Sep 2026
Instances.eventAt	propext, Quot.sound	26 Sep 2026
Instances.eventCarried	propext, Quot.sound	26 Sep 2026
Instances.eventRepair	propext, Quot.sound	26 Sep 2026
Instances.eventTrajectory	propext, Quot.sound	26 Sep 2026
Instances.eventTransition	none	26 Sep 2026
Instances.eventE3Reinstatement	propext, Quot.sound	26 Sep 2026
Instances.eventHistoryAgrees	propext, Quot.sound	26 Sep 2026
Instances.eventFullyHistoryLinked	propext, Quot.sound	26 Sep 2026
Instances.eventExecutedReinstatement	propext, Quot.sound	26 Sep 2026
Instances.noEpisodeLinkedWithEmptyHistory	propext	26 Sep 2026
Instances.declaredRecordList_nodup	propext, Quot.sound	26 Sep 2026
Instances.declaredRecordList_subset	propext, Quot.sound	26 Sep 2026
Instances.declaredRecordList_length	propext	26 Sep 2026
Instances.declaredCapacityBound	propext, Classical.choice, Quot.sound	26 Sep 2026
Instances.declaredTowerCapacityBound	propext, Classical.choice, Quot.sound	26 Sep 2026
Instances.copiedRecordReindexNotDeclared	none	26 Sep 2026
Instances.eventCheckedSystemRepair	propext, Quot.sound	26 Sep 2026
Instances.eventLowerInspectionPasses	propext, Quot.sound	26 Sep 2026
Instances.eventUpperRecordedCheck	propext, Quot.sound	26 Sep 2026
Instances.eventCheckedTowerAuditsLowerStack	propext, Quot.sound	26 Sep 2026
Instances.eventCheckedLevelsDisjoint	propext, Quot.sound	26 Sep 2026
Instances.eventCheckedConcreteCapacity	propext, Quot.sound	26 Sep 2026
Instances.eventCheckedCapacityBound	propext, Classical.choice, Quot.sound	26 Sep 2026
Instances.eventCheckedStatusSaturated	propext, Quot.sound	26 Sep 2026
Instances.eventCheckedExecutedRepair	propext, Quot.sound	26 Sep 2026
E2_SelfSoundnessObstruction	propext	2 Sep 2026
E2_CapacityBound	propext, Classical.choice, Quot.sound	2 Sep 2026

Declaration	Axioms	Record
Instances.noE2WithOccurringLevel	propext, Classical.choice, Quot.sound	26 Sep 2026
Instances.spikeNoE2	propext, Classical.choice, Quot.sound	26 Sep 2026
E6_AttentionKKT	propext, Classical.choice, Quot.sound	2 Sep 2026
E6_SlackCollapse	propext, Classical.choice, Quot.sound	2 Sep 2026
Instances.weightedSaturatedSupportingPlane	propext, Classical.choice, Quot.sound	26 Sep 2026
Instances.capMultiplierKKT_sufficient	propext, Classical.choice, Quot.sound	26 Sep 2026
Instances.capExampleSlopeZero	propext, Classical.choice, Quot.sound	26 Sep 2026
Instances.capExampleGloballyOptimal	propext, Classical.choice, Quot.sound	26 Sep 2026
Instances.unitCapExampleGloballyOptimal	propext, Classical.choice, Quot.sound	26 Sep 2026
Instances.actualFamilyStrictExtensions	none	26 Sep 2026
Instances.clockedOddRepair	propext, Quot.sound	26 Sep 2026
Instances.clockedEvenPerturbation	propext, Quot.sound	26 Sep 2026
Instances.clockedFamilyEveryEntryInstalled	propext, Quot.sound	26 Sep 2026
Instances.clockedUnbounded	propext, Quot.sound	26 Sep 2026
Instances.periodicNovel	propext, Quot.sound	26 Sep 2026
Instances.periodicStrictAt	propext, Classical.choice, Quot.sound	26 Sep 2026
Instances.clockedStrictExtensions	propext, Classical.choice, Quot.sound	26 Sep 2026
Instances.periodicRefinementsDistinct	propext	26 Sep 2026
E1_StrictSelfExtension	none	2 Sep 2026
Instances.finiteForcingNoPerpetualNovelty	none	26 Sep 2026
Instances.finiteForcingUninhabited	Classical.choice	26 Sep 2026
Instances.E1_StrictSelfExtension_noCertificate	Classical.choice	26 Sep 2026
Instances.E12Semantic.repairClosed_inter	propext, Quot.sound	26 Sep 2026
Instances.E12Semantic.integrated_disjoint_or_equal	propext, Classical.choice, Quot.sound	26 Sep 2026
Instances.E12Semantic.repairPart_integrated	propext, Quot.sound	26 Sep 2026
Instances.E12Semantic.repairedIndividualWitness	propext, Quot.sound	26 Sep 2026
Instances.E12Semantic.healthyOnly_closure_loadBearing	propext, Quot.sound	26 Sep 2026
Instances.E12Semantic.secondLeft_maximal	propext, Quot.sound	26 Sep 2026
Instances.E12Semantic.secondRight_maximal	propext, Quot.sound	26 Sep 2026
Instances.E12Semantic.second_no_integrated	propext, Quot.sound	26 Sep 2026
Instances.E13Semantic.compose_payment	propext, Quot.sound	26 Sep 2026
Instances.E13Semantic.chain_obstruction_bound	propext, Quot.sound	26 Sep 2026
Instances.E13Semantic.chain_length_bound	propext, Quot.sound	26 Sep 2026
Instances.E13Semantic.cross_context_zero_to_one	propext, Quot.sound	26 Sep 2026
Instances.E13Semantic.cross_context_one_to_zero	propext, Quot.sound	26 Sep 2026
Instances.E13Semantic.cross_context_two_step_bound	propext, Quot.sound	26 Sep 2026
Instances.E13Semantic.concrete_composite_payment	propext, Quot.sound	26 Sep 2026
Instances.E13Semantic.concrete_composite_reduces	propext, Quot.sound	26 Sep 2026
Instances.E13Semantic.roleOnly_mapsRepair	none	26 Sep 2026
Instances.E13Semantic.roleOnly_no_reduction	propext, Quot.sound	26 Sep 2026
E9_CuriosityArgmax	propext, Classical.choice, Quot.sound	2 Sep 2026
E6_E9_AccessArbitration	propext, Classical.choice, Quot.sound	2 Sep 2026
E6_E9_AccessArbitration_epsilon	propext, Classical.choice, Quot.sound	2 Sep 2026
selected_access_move_ratio_bound	propext, Classical.choice, Quot.sound	2 Sep 2026
selected_access_move_ratio_bound_epsilon	propext, Classical.choice, Quot.sound	2 Sep 2026
E9_SameFamilyStrictness	propext, Classical.choice, Quot.sound	2 Sep 2026
Instances.offlineCapacityFromExhaustion	propext, Classical.choice, Quot.sound	26 Sep 2026

Declaration	Axioms	Record
Instances.geometryCapacityFromExhaustion	propext, Classical.choice, Quot.sound	26 Sep 2026
E4_Brittleness	none	2 Sep 2026
E1_StatusPartition	none	2 Sep 2026
E1_Internalization	none	2 Sep 2026
E1_NoGeneratorDichotomy	none	2 Sep 2026
E3_TwoLevelFixedPoint	propext, Classical.choice, Quot.sound	2 Sep 2026
E3_StatusPartition	propext, Classical.choice, Quot.sound	2 Sep 2026
E3_RegressStoppedByE2	propext, Classical.choice, Quot.sound	2 Sep 2026
E3_AblationSeparation	propext, Classical.choice, Quot.sound	2 Sep 2026
compilationLawful_requiredCoreGatesPass	propext	2 Sep 2026
E4_Compilation	propext, Quot.sound	2 Sep 2026
E4_StatusPartition	propext	2 Sep 2026
E5_CollapseFalsifier	none	2 Sep 2026
E5_ReclosureCollapse	none	2 Sep 2026
E5_RevivedExcludesLowerPriority	none	2 Sep 2026
E5_IrreversibleCollapse	none	2 Sep 2026
E5_SubsidyWithdrawalReclassification	none	2 Sep 2026
E7_AlarmTrichotomy	propext, Classical.choice, Quot.sound	2 Sep 2026
E7_PreemptionSignature	none	2 Sep 2026
E8_ControlPrice	propext, Classical.choice, Quot.sound	2 Sep 2026
E8_ComponentShadowPrice	propext, Classical.choice, Quot.sound	2 Sep 2026
E8_CompressedSummaryLawfulness	propext, Classical.choice, Quot.sound	2 Sep 2026
E8_1_AuditCapture	propext, Classical.choice, Quot.sound	2 Sep 2026
E8_1_CaptureFalsifier	propext, Classical.choice, Quot.sound	2 Sep 2026
E8_1_MetaAuditBoundedByE2	propext, Classical.choice, Quot.sound	2 Sep 2026
BudgetInflationAudit.lineageMustMatchClassifiedMove	propext, Classical.choice, Quot.sound	2 Sep 2026
E10_CognitiveDemarcation	propext	2 Sep 2026
E10_ScheduleTrapNull	none	2 Sep 2026
E10_ScheduleTrapExcludesLowerPriority	propext	2 Sep 2026
E10_DecodableCorrelate	none	2 Sep 2026
E10_1_Intention	propext	2 Sep 2026
E10_1_PostHocIntentionFalsifier	none	2 Sep 2026
E10_1_PostHocIntentionExcludesLowerPriority	propext	2 Sep 2026
E10_2_Goal	propext	2 Sep 2026
E10_2_RewardProxyNotGoal	none	2 Sep 2026
E10_2_RewardProxyOnlyExcludesLowerPriority	propext	2 Sep 2026
E11_StackActivity	none	2 Sep 2026
E11_Constitutive	none	2 Sep 2026
E11_ConditioningOnly	none	2 Sep 2026
E11_WashoutInert	none	2 Sep 2026
E11_NCTDObstruction	propext	2 Sep 2026
E11_StatusPartition	none	2 Sep 2026
E12_Individuation	none	2 Sep 2026
E12_StatusPartition	none	2 Sep 2026
Instances.E12_singleton_maximal_from_tests	propext	26 Sep 2026
Instances.spikeSelfMaintained	propext, Classical.choice, Quot.sound	26 Sep 2026
E13_Communication	none	2 Sep 2026
E13_TeachingCapacity	none	2 Sep 2026
E13_CoercionNull	none	2 Sep 2026
E13_SymbolicRepair	none	2 Sep 2026
E13_StatusPartition	propext	2 Sep 2026
E13_CoercionNullExcludesLowerPriority	propext	2 Sep 2026

Declaration	Axioms	Record
Instances.E13_two_contexts_distinct	none	26 Sep 2026
Instances.E13_two_contexts_members	propext	26 Sep 2026
E14_ReconsolidationStatus	propext, Quot.sound	2 Sep 2026
E14_LawfulConflictTrichotomy	propext, Quot.sound	2 Sep 2026
E14_OutcomeCollisionExcludesLawful	propext, Quot.sound	2 Sep 2026
E14_RecordRepair	propext, Quot.sound	2 Sep 2026
E14_RecordCoarsening	propext, Quot.sound	2 Sep 2026
E14_StatusedUnresolvedChargedToLambda	propext, Quot.sound	2 Sep 2026
E14_SilentRewriteFalsifier	propext, Quot.sound	2 Sep 2026
E14_RetrievalWithoutTransportControl	propext, Quot.sound	2 Sep 2026
E14_LaunderedProvenanceDefect	none	2 Sep 2026
E15_OfflineReclosureStatus	propext, Classical.choice, Quot.sound	2 Sep 2026
E15_DeficitAlternation	propext, Classical.choice, Quot.sound	2 Sep 2026
E15_OnlineSufficientNoOfflineRequired	propext, Classical.choice, Quot.sound	2 Sep 2026
E15_OfflinePermittedOutsideDeficit	propext, Classical.choice, Quot.sound	2 Sep 2026
E15_SkippedOfflineCascade	propext, Classical.choice, Quot.sound	2 Sep 2026
E15_DecorativeOfflineFalsifier	propext, Classical.choice, Quot.sound	2 Sep 2026
E15_DutyCycleMismatchFalsifier	propext, Classical.choice, Quot.sound	2 Sep 2026
E15_DeficitOnlineFalsifiesNecessity	propext, Classical.choice, Quot.sound	2 Sep 2026
E15_DeficitCounterexampleExcludesLowerPriority	propext, Classical.choice, Quot.sound	2 Sep 2026
E15_OffInventoryFlowCannotCertifyOffline	propext, Classical.choice, Quot.sound	2 Sep 2026
E16_AdaptabilityStatus	propext, Classical.choice, Quot.sound	2 Sep 2026
E16_CoherentAdaptability	propext, Classical.choice, Quot.sound	2 Sep 2026
E16_CurrentAuditBlindness	propext, Classical.choice, Quot.sound	2 Sep 2026
E16_RouteManufacturesRepairCapacity	propext, Classical.choice, Quot.sound	2 Sep 2026
E16_ArtifactNotAdaptability	propext, Classical.choice, Quot.sound	2 Sep 2026
E16_FlatNoAdaptability	propext, Classical.choice, Quot.sound	2 Sep 2026
E16_FlattenableNotAdaptability	propext, Classical.choice, Quot.sound	2 Sep 2026
E16_CurrentizableIsLegibleSlack	propext, Classical.choice, Quot.sound	2 Sep 2026
E16_DissipativeNotAdaptability	propext, Classical.choice, Quot.sound	2 Sep 2026
E16_BoundedPerturbationFailureRejected	propext, Classical.choice, Quot.sound	2 Sep 2026
E16_SupportConfoundExcludesLowerPriority	propext, Classical.choice, Quot.sound	2 Sep 2026
E16_LoopAsymmetryAnchor	Quot.sound	2 Sep 2026
E16_SwappedRouteTrialPopulationRejected	propext, Quot.sound	2 Sep 2026
E16_FlatSecondArmUsesClaimLinkedPair	propext, Classical.choice, Quot.sound	2 Sep 2026
E16_UnrelatedEqualPairCannotReplaceClaimDifference	propext, Classical.choice, Quot.sound	2 Sep 2026

S9.3 Hypotheses carried but not used

Several Lean statements of the Templates carry a hypothesis that their proof does not use; each Lean statement is then a special case of what its proof establishes. The cases are: in E1, closed-loop scope and finite forcing in the internalization classification; in E2, the audited-repair trigger of forced stratification and the ordered tower listing in the declared bound; in E6, the binding budget of Theorem 4.9; in E9 and the arbitration statements, positive costs; in E4, the window’s obstruction reduction and idempotence stability, and $C' \neq C$ in Proposition 5.4; in E5, the prior status and the withdrawal event; and in E7, the blind-spot, coupling, policy and active-family hypotheses. The printed statements omit these hypotheses.

S9.4 What each Lean statement covers

Table S20 records, for each entry, which parts of the printed statement the Lean declarations establish and which are argued on paper, imported or supplied as certificates.

Table S20. Printed statements and their Lean coverage.

Entry	In Lean	On paper, imported or supplied
D1	all semilattice clauses	admissibility of joint access (not claimed)
D2	the rational clauses; trace nonnegativity and monotonicity	the chain rule and positive semidefiniteness (hypotheses, imported)
D3–D6	the consistency lemmas	completeness declarations (supplied)
E1	forced self-extension from an actual-family certificate; the clocked device; exclusivity of five statuses; the endogenous branch’s fields; the no-generator case; the version-1 certificate is uninhabited	the complete status record; the certified inputs
E2	the declared capacity bound and anti-reindexing; the audited two-level tower on the executed device; the priority status partition; the vacuity of the version-1 measure; the self-soundness obstruction in weak form	forced stratification (tag order and audit relation supplied)
E3	executed repair and history-linked reinstatement for the device, and failure without episodes; exclusivity of four statuses; packaging of the ablation curves	the maintained-status record; measured curves; regress stop (first-version measure)
E4	compiled status from a full case; exclusivity of five statuses; brittleness from a split pair	the compilation case; attribution of the compiled operator to the window
E5	falsifier refutes collapse; exclusivity of seven statuses; reclassification after withdrawal	the rescue inventory; viability kernel and reachability records
E6	multiplier algebra; slack collapse; sufficiency for capped linear objectives; the numerical optimum at a cap	sufficiency under concavity (argued on paper); Example 4.13 and Remark 4.14; an instance on a carried probe economy (open)
E7	exclusivity of three dispositions; preemption signature	the disposition package; alarm fatigue (on paper)
E8	component and summary consequences; exclusivity of seven statuses; capture and its falsifier	the envelope derivative (on paper); the meta-audit bound (first-version measure)
E9	finite argmax; arbitration; selected-move bound; operator saturation	identification of the discharge with the operator contraction

Entry	In Lean	On paper, imported or supplied
E10	exclusivity of eight, five and four statuses; positive evidence forms; exclusion statements	the status records and comparators
E11	exclusivity of four statuses; conditioning and washout; the blocked-channel witness	the rewrite comparison; a bridge from E4 (open)
E12	intersection of repair-closed parts; disjointness of integrated individuals; the three-state individual and the overlapping-maxima example; integrated status from a case; exclusivity of six statuses; the singleton candidate	budget and record closure and the coarsest viability boundary (certificate); operational selfhood (on paper)
E13	composition of repair transports; the chain bound; the cross-context instance and the flat counterexample; the four judgments from full cases; coercion-null priority	the full communication, teaching, coercion and symbolic certificates
E14	lawful trichotomy; nine statuses; exclusion statements	the outcome inventories
E15	exclusivity of eight statuses; the reallocation identity derived from exhaustion; off-inventory flows	the alternation case
E16	exclusivity of eight statuses; current-event equivalence and unequal probabilities from the record; regression statements	the status record; any causal reading (not claimed)

S10 Replay

All commands run from the repository root.

Lean library.

```
cd lean
lake build
```

The default targets build the library, the vendored interfaces and the instance modules. The axiom transcript of the instance theorems in the concordance is reproduced by

```
cd lean
lake env lean ../paper/supplement/instance_axioms.lean
```

Finite computations.

```
cd lab
pip install -e '[dev]'
pytest -q
PYTHONPATH=. python -m sixbirds_foundations_v.sweeps.e16_adaptability_sweep
```

The test suite runs the computations of E1–E15; the E16 computation runs separately.

Supplement tables and papers.

```
python3 paper/supplement/build_supp.py
cd paper
latexmk -pdf -interaction=nonstopmode -halt-on-error -outdir=build main.tex
latexmk -pdf -interaction=nonstopmode -halt-on-error -outdir=build supplement.tex
latexmk -pdf -interaction=nonstopmode -halt-on-error -outdir=build main.tex
```

The main paper and the supplement cite each other's numbered items, so the main paper is built a second time after the supplement.

References

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